Bounded-list decoding radius inheritance for quotient Reed-Muller codes

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doi:10.4230/LIPIcs.CCC.2025.1; CC BY4.0

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Editorial note. Selected complete native author theorem, proof, full supporting sections and bibliography in standalone layout. TeX input files flattened as explicit spans. Unused frontmatter/packages omitted. No mathematical corrections or new proofs.

Allowed standard prerequisites (editorial).

- Finite-field Reed-Muller distance and Bhowmick-Lovett low-complexity approximation/degree preservation. Full author preliminaries and exact cited standard guarantees included.

2 Preliminaries

2.1 Basic Definitions and Notations

We denote by $\mathbb{N}$ the set set of integers, i.e natural numbers (excluding 0). For an integer k we denote $[k] \coloneqq \{1, 2, ..., k\}$. We use $y = x \pm \epsilon$ to denote $y \in [x - \epsilon, x + \epsilon]$, and similarly $y = x \mp \lambda$ to denote $y \in [x + \lambda, x - \lambda]$ (usually when $\lambda < 0$).
Fix a prime field $\mathbb{F} = \mathbb{F}_p$. Denote by $|\cdot|$ the natural map of $\mathbb{F}$ to $\{1, ..., p-1\} \in \mathbb{N}$. We denote the character from $\mathbb{F}$ by $e[x] \coloneqq e^{2\pi i \cdot |x|}$

Generally speaking and unless stated otherwise, we use the following conventions: We use $n \in \mathbb{N}$ to denote the number of variables in Reed-Muller code. We use d to denote a degree (typically the degree of the polynomials in our code), and $\tilde{X}$ to denote the subset of $\mathbb{F}^n$ we work in i.e $\tilde{X} \subseteq \mathbb{F}^n$. Properties of the subset $\tilde{X}$ will usually be denoted with $\tilde{\square}$. We use F, G, H to denote general functions with domain $\mathbb{F}^n$, and f, g, h to denote functions with domain $\tilde{X}$. We use $\mathfrak{F}, \mathfrak{G}, \mathfrak{H}$ and $\mathfrak{f}, \mathfrak{g}, \mathfrak{h}$ respectively to denote sets of such functions. Similarly, we use P, Q, H to denote polynomials with domain $\mathbb{F}^n$, and p, q, h polynomials with domain $\tilde{X}$. We use $\mathcal{P}, \mathcal{Q}, \mathcal{H}$ and $\mathfrak{p}, \mathfrak{q}, \mathfrak{h}$ respectively to denote sets of such polynomials.

2.2 Polynomials in $\mathbb{F}^n$

We start by presenting a standard definition for a polynomial over a finite field.

Definition 2.1 (Polynomial: Global Definition)**.** *Let $d \in \mathbb{N}$ be a constant. A function $P : \mathbb{F}^n \to \mathbb{F}$ is called a* polynomial of degree $\leq d$ *if it is of the following form:*

$$P(x_1, ..., x_n) = \sum_{0 \leq d_1, ..., d_n : \sum_{i=1}^n d_i \leq d} c_{d_1,...,d_n} \prod_{i=1}^n x_i^{d_i}$$

We denote the set of all polynomials of degree $\leq d$ by $Poly_{\leq d}(\mathbb{F}^n \to \mathbb{F})$. The value d in the definition above is called the global degree *of the function P, shorthand by the* degree *of P, and it is denoted by* $\deg(P) = d$.
Additionally, the set of all polynomials from $\mathbb{F}^n$ to $\mathbb{F}$ of degree $\leq d$ is denoted by:

$$Poly_{\leq d}(\mathbb{F}^n \to \mathbb{F})$$

Next, we present a known equivalent definition for a polynomial using derivatives. To do so, we first define a derivative in the case of finite fields.

Definition 2.2 (Derivative)**.** *Given a function $F : \mathbb{F}^n \to \mathbb{F}$ and $a \in \mathbb{F}^n$, we define the derivative of F in direction a as a function $D_a F : \mathbb{F}^n \to \mathbb{F}$ defined as follows:*

$$D_a F(x) \coloneqq F(x + a) - F(x)$$

Lemma 2.3. *Let $d \in \mathbb{N}$. A function $F : \mathbb{F}^n \to \mathbb{F}$ is a polynomial of degree $\leq d$ if and only if $D_a F$ is a polynomial of degree $\leq d - 1$ for all $a \in \mathbb{F}^n$.*

This leads us to a natural definition of a *degree* of a function using derivatives.

Definition 2.4 (Local Degree)**.** *For a function $F : \mathbb{F}^n \to \mathbb{F}$, we define its* local degree*, to be the least integer $d \in \mathbb{N}$ such that for all $a_1, ..., a_{d+1}, x \in \mathbb{F}^n$:*

$$D_{a_{d+1}} ... D_{a_1} F(x) = 0$$

In $\mathbb{F}^n$, the two definitions of degree coincide, and we get a single definition of a degree:

Lemma 2.5 (Equivalance of definitions of a degree)**.** *Let $F : \mathbb{F}^n \to \mathbb{F}$ be a function, and let $d \in \mathbb{N}$ be an integer. Then, the global degree of a F* equals *its local degree.*

Remark 2.6. *We sometimes refer to the requirement that the local degree of a function is $\leq d$, as the* local criteria *of degree $\leq d$ polynomials.*

2.3 Rank-bias in $\mathbb{F}^n$

We start by defining the notion of *bias*, which is a measure of how the function is far from being equidistributed (see Appendix 7 for the exact details).

Definition 2.7 (Bias)**.** *Let $F : \mathbb{F}^n \to \mathbb{F}$. The* bias *of the function F is defined in the following way:*

$$bias(F) \coloneqq 1/\left|\mathbb{F}^n\right| \cdot \sum_{x \in \mathbb{F}^n} e\left[F(x)\right]$$

Moreover, for a subset $\tilde{X} \subseteq \mathbb{F}^n$, we define the bias of F in $\tilde{X}$ *to be:*

$$bias_{\tilde{X}}(F) \coloneqq 1/\left|\tilde{X}\right| \cdot \sum_{x \in \tilde{X}} e\left[F(x)\right]$$

Next, we present a standard definition of rank of a polynomial, which is a notion that measures how *structured* is the function. Note that low rank implies the polynomial is highly structured. Formally we have the following definition:

Definition 2.8 (Rank of a Polynomial)**.** *Given a constant $d \in \mathbb{N}$ and a polynomial P, the d-*rank *of P, denoted as $rank_d(P)$ is defined to be the smallest integer r such that P can be computed given r polynomials of degree $< d$. In other wards, we say $rank_d(P) = r$ if r is the smallest integer such that there exists r polynomials $Q_1, ..., Q_r \in Poly_{\leq d-1}(\mathbb{F}^n \to \mathbb{F})$ and a function $\Gamma : \mathbb{F}^n \to \mathbb{F}$ such that:*

$$P(x) = \Gamma\left(Q_1(x), ..., Q_r(x)\right)$$

If $d = 1$, then 1-rank is defined to be ∞ if P is non constant, and 0 otherwise. Moreover, for a polynomial P of degree $\deg(P) = d$ we denote $rank(P) \coloneqq rank_d(P)$.
We call such function Γ a decomposition *or a* computation *of P using lower-degree polynomials.*

Let us now define a factor. Note that we focus our discussion to factors in $\mathbb{F}^n$, but define the basic definitions over a general set U so they will apply for factors over a general sets. This is necessary as we will later use them also for other sets such as $\tilde{X} \subseteq \mathbb{F}^n$.

Definition 2.9 (Factor)**.** *Let U be a set. A* factor over U*, denoted by $\mathcal{B}$, is simply a partition of the set U. Each subset in the partition is called an* atom*. A collection of function $F_1, ..., F_c : U \to \mathbb{F}$ defines a factor $\mathcal{B}_{F_1,...,F_c}$ over U with atoms:*

$$\{u \in U | f_1(u) = b_1, ..., f_c(u) = b_c\}$$

for all $b_1, ..., b_c \in \mathbb{F}$.
Additionally, we use $\mathcal{B}$ to also denote the map $\mathcal{B}(u) \to (F_1(u), ..., F_c(u))$.

Notation. *Let $F_1, ..., F_c : U \to \mathbb{F}$ be a collection of functions. For a factor $\mathcal{B} \coloneqq \mathcal{B}_{F_1,...,F_c}$, we denote by $|\mathcal{B}|$ the amount of functions that define it, i.e. $|\mathcal{B}| = c$. Moreover, we denote $\|\mathcal{B}\| \coloneqq |\mathbb{F}|^c$, which is the maximal amount of (possibly empty) atoms. Additionally, the rank of the factor is defined to be the rank of the polynomials that generate it.*

Definition 2.10 (Polynomial Factor)**.** *We say a factor $\mathcal{B}$ over $\mathbb{F}^n$ is a* polynomial factor *if it is defined by a collection of polynomials $P_1, ..., P_c : \mathbb{F}^n \to \mathbb{F}$, i.e. $\mathcal{B} = \mathcal{B}_{P_1,...,P_c}$. The degree of the factor, denote as $\deg(\mathcal{B})$ is the maximal degree of the polynomials $P_1, ..., P_c$.*

Note that the notion of degree (and polynomial) are defined only for functions over $\mathbb{F}^n$, therefore this definition is well-defined only for $U = \mathbb{F}^n$.

Definition 2.11 (Rank of a Factor)**.** *Let $\mathcal{P}$ be a collection of polynomials $P_1, ..., P_c : \mathbb{F}^n \to \mathbb{F}$. The rank of the polynomial collection is defined as:*

$$rank\left(\mathcal{P}\right) \coloneqq \min\left\{ rank_d\left(\sum_{i=1}^{c} \lambda_i P_i\right) \middle| 0 \neq \vec{\lambda} \in \mathbb{F}^c, d = \max_{i\in[c]} \deg(\lambda_i P_i) \right\}$$

For a factor $\mathcal{B}$ defined by a collection of polynomials $\mathcal{P}$, we define its rank to be the rank of the collection of polynomials defining it. For a non-decreasing function $r : \mathbb{N} \to \mathbb{N}$, a factor $\mathcal{B}$ is called r-regular if its rank is at least $r(|\mathcal{B}|)$.

Note. *Note that in the definition above, the rank of each linear combination is calculated as the d-rank, where d is the maximal degree of a polynomial that participates in the linear combination non-trivially. This is crucial as it ensures that a high rank factor do not have linear dependence in the largest-degree homogenous component of any of its polynomials.*

We now present a fundamental property of high rank polynomials, that was first proved by [GT07a] when $d < |\mathbb{F}|$, later extended to general fields by [KL08], and further extended also to large fields by [BL15]. This property of high rank polynomials is that they they have low bias:

Theorem 2.12 (Rank-bias in $\mathbb{F}^n$)**.** *Let $\mathbb{F}$ be a finite field. Let $\epsilon > 0$ and $d \in \mathbb{N}$. There exists $r_{2.12} \coloneqq r_{2.12}(\mathbb{F}, d, \epsilon)$, such that for every degree-d polynomial $P : \mathbb{F}^n \to \mathbb{F}$: if $rank\left(P\right) \geq r_{2.12}$ then $bias(P) < \epsilon$.*

Remark 2.13. *This property implies that a collection of polynomials that have high rank is* equidistributed. *See Appendix 7 for more details in this regard.*

2.4 Regularization in $\mathbb{F}^n$

In this subsection we define the regularization process in $\mathbb{F}^n$. Before doing so, let us present some definitions in this regard. Note that we define the basic definitions over a general set U so they will apply for factors over a general sets, as this is necessary as we will later use them also for other sets such as $\tilde{X} \subseteq \mathbb{F}^n$.

Definition 2.14 (Measureable)**.** *Let U be a set, and let $A \subseteq U$. Let $\mathfrak{F} = \{F_1, ..., F_c\}$ be a collection of functions $F_i : U \to \mathbb{F}$. We say a function $G : U \to \mathbb{F}$ is* measurable in respect of $\mathfrak{F}$ in A*, shorthand by* $\mathfrak{F}$-measurable in A*, if there exists a function $\Gamma : \mathbb{F}^c \to \mathbb{F}$ such that:*

$$\forall a \in A : g(a) = \Gamma(F_1(a), ..., F_c(a))$$

When discussing the factor over A defined by $\mathcal{B} = \mathcal{B}_{F_1,...,F_c}$, we also say G is measurable in resepct of $\mathcal{B}$*. The function Γ will be denoted as the* measurement function *of G in respect of $\mathfrak{F}$. Additionally, when $A = U$, we sometimes omit the specification of the domain, and say G is measurable in respect of $\mathfrak{F}$.*
Note that in this paper, we usually think of $U = \mathbb{F}^n$, and A is either $\mathbb{F}^n$ or $\tilde{X} \subseteq \mathbb{F}^n$.

Remark 2.15. *If G is $\mathfrak{F}$-measurable in A, then every value of G in A can be determined by the values of $F_1, ..., F_c$. In other words, the function G is constant inside every atom of the factor defined by $\mathfrak{F}$.*

Definition 2.16 (Syntactic Refinement)**.** *Let $\mathcal{B}$ and $\mathcal{B}'$ be polynomial factors over U. We say a factor $\mathcal{B}'$ is a* syntactic refinement *of the factor $\mathcal{B}$, if the collection of functions defining $\mathcal{B}$ is a subset of the set of functions defining $\mathcal{B}'$. We denote this property of $\mathcal{B}'$ by $\mathcal{B}' \succeq_{syn} \mathcal{B}$.*

We now present a standard generalized definition of refinement, where we only require the atoms induced by the refined factors are sub-atoms of those that are induced by the original factor. Note that in this refinement, we allow the refined factor to include completely different polynomials than the original factor.

Definition 2.17 (Semantic Refinement)**.** *Let $\mathcal{B}$ and $\mathcal{B}'$ be polynomial factors on U defined by $\mathcal{P}$ and $\mathcal{P}'$ respectively. We say the factor $\mathcal{B}'$ is a* semantic refinement *of the factor $\mathcal{B}$ in $A \subseteq U$, if $x, y \in A$ with $\mathcal{B}'(x) = \mathcal{B}'(y)$ implies that $\mathcal{B}(x) = \mathcal{B}(y)$. We denote this property of $\mathcal{B}'$ by $\mathcal{B}' \succeq_{sem|A} \mathcal{B}$. When $A = U$, we sometimes omit A from the syntax and denote it with $\mathcal{B}' \succeq_{sem} \mathcal{B}$*
Note that $\mathcal{B}' \succeq_{syn} \mathcal{B}$ implies $\mathcal{B}' \succeq_{sem|A} \mathcal{B}$ for every $A \subseteq U$.

Remark 2.18. *A handy property of semantic refinement is that if $F : A \to \mathbb{F}$ is $\mathcal{P}$-measurable, then it is also $\mathcal{P}'$-measurable in A. Moreover, the other direction is also true: If every $\mathcal{P}$-measurable function $F : A \to \mathbb{F}$ in A is also $\mathcal{P}'$-measurable in A, then $\mathcal{B}' \succeq_{sem|A} \mathcal{B}$.*

Next, we recall a lemma that was presented in [BFH+13, Theorem 4.1], that allows us, given a polynomial that is measurable by a high rank factor in $\mathbb{F}^n$, to replace the polynomials in the measurement function to any collection of polynomials with smaller or equal degree, and preserve the degree of the original polynomial. Note that we state the lemma under the constraint that $d < |\mathbb{F}|$, but it is also valid for when $d \geq |\mathbb{F}|$ with proper generalization of definitions to this case (See [BFH+13, Theorem 4.1] for the exact statement).

Lemma 2.19 (Preserving Degree in $\mathbb{F}^n$)**.** *Let $d > 0$ an integer such that $d < |\mathbb{F}|$, and let $P_1, ..., P_c : \mathbb{F}^n \to \mathbb{F}$ be polynomials of degree at most d, that form a factor of rank $\geq r^{2.19}(\mathbb{F}, d, c)$. Assume that for $\Gamma : \mathbb{F}^c \to \mathbb{F}$, the function $\gamma : \mathbb{F}^n \to \mathbb{F}$ defined as $\gamma(a) \coloneqq \Gamma(P_1(a), ..., P_c(a))$ is of $\deg(\gamma) = d'$.*
Then, for every collection of polynomials $Q_1, ..., Q_c : \mathbb{F}^n \to \mathbb{F}$ that satisfy $\deg(Q_i) \leq \deg(P_i)$, the function γ' defined as $\gamma'(a) = \Gamma(Q_1(a), ..., Q_c(a))$ is a polynomial of $\deg(\gamma') \leq d'$.

Next, we restate a useful lemma from [BL15, Lemma 4.17] that shows that under the conditions above, Γ is as a low-degree polynomial (with even stronger conditions). Formally, they showed:

Lemma 2.20 (Faithful Composition)**.** *In the case discussed above, the structure of Γ is as follows:*

$$\Gamma(z_1, ..., z_{c_1}) = \sum_{\alpha \in [p-1]^c} C_\alpha \cdot \prod_{i=1}^{c} z_i^{\alpha_i}$$

where $C_\alpha = 0$ whenever $\sum_{i=1}^{c_1}(\alpha_i \cdot \deg(P_i)) > d'$.
In other words, this means that Γ as a function $\Gamma : \mathbb{F}^c \to \mathbb{F}$, is a polynomial

of degree $\leq d'$, even when substituting its i-th input by any polynomial of degree $\leq \deg(P_i)$.

Finally, we restate the regularization process, that was first presented by [GT07a, Lemma 2.3]. The regularization process shows that every factor have a high-rank factor that semantically refines it, without increasing the size of the factor too much (its new size is independent of n).

Lemma 2.21 (Regularization in $\mathbb{F}^n$)**.** *Let $r : \mathbb{N} \to \mathbb{N}$ be a non-decreasing function and let $d \in \mathbb{N}$. There exists $C^{2.21}_{r,d} : \mathbb{N} \to \mathbb{N}$ such that the following holds: Let $\mathcal{B}$ be a factor on $\mathbb{F}^n$ defined by polynomials $\mathcal{P} = (P_1, ..., P_c)$ where for all $i \in [c]$: $P_i : \mathbb{F}^n \to \mathbb{F}$ and $\deg(P_i) \leq d$. Then, there is an r-regular factor $\mathcal{B}'$ defined by polynomials $\mathcal{Q} = (Q_1, ..., Q_{c'})$ where for all $i \in [c]$: $Q_i : \mathbb{F}^n \to \mathbb{F}$ and $\deg(Q_i) \leq d$ such that $\mathcal{B}' \succeq_{sem} \mathcal{B}$ and $c' \leq C^{2.21}_{r,d}(c)$.*
Moreover, if $\mathcal{B} \succeq_{syn} \bar{\mathcal{B}}$ for some polynomial factor $\bar{\mathcal{B}}$ with rank at least $r(c') + c' + 1$, then we can require that $\mathcal{B}' \succeq_{syn} \bar{\mathcal{B}}$.

3 Polynomials in $\tilde{X}$

In this section we wish to generalize the definition of degree-d polynomials for functions $f : \tilde{X} \to \mathbb{F}$. Note that we wish to define it using a property of f that is intrinsic to $\tilde{X}$: given a function $f : \tilde{X} \to \mathbb{F}$, we wish be able to determine its degree only using values of $\tilde{X}$, without considering any value outside of $\tilde{X}$ (such as values of $\mathbb{F}^n \setminus \tilde{X}$).
To define such property, we generalize the local definition of a degree that is defined for polynomials in $\mathbb{F}^n$. We remind the reader that in $\mathbb{F}^n$, we said a function over $\mathbb{F}^n$ is a polynomial of degree $\leq d$ if and only if its $(d+1)$-derivative in every direction is $\equiv 0$. Thus, in order to determine the $(d+1)$-derivative of a function in directions $y_1, ..., y_{d+1}$, one needs to evaluate the function over all the points of the cube generated by $x, y_1, ..., y_{d+1}$, which is the set of points $\left\{x + \sum_{i \in S} y_i\right\}_{S \subseteq [d+1]}$. This raises a challnge in extending this definition for functions defined over $\tilde{X} \subseteq \mathbb{F}^n$: depending on $\tilde{X}$, the function $f : \tilde{X} \to \mathbb{F}$ is not be defined to all points in all the cubes of $\mathbb{F}^n$, because some of those points do not lie in $\tilde{X}$.
Therefore, to generalize the definition of a polynomial to $\tilde{X}$, we start by giving the formal definition and notation of the set of cubes in $\tilde{X}$:

Definition 3.1 (Cubes)**.** *Let $k \in \mathbb{N}$ be an integer and let $x, y_1, ..., y_k \in \mathbb{F}^n$. We define the cube $(x|y_1, ...y_k)$ as follows:*

$$(x|y_1, ..., y_k) \coloneqq \left\{ x + \sum_{i \in S} y_i \right\}_{S \subseteq [k]}$$

We refer to x as the offset *of the cube, and $y_1, ..., y_k$ as the* directions *of the cube.*
Moreover, Let $\tilde{X} \subseteq \mathbb{F}^n$ be a subset. We define the set of cubes of $\tilde{X}$ of size k *as follows:*

$$C_k(\tilde{X}) \coloneqq \left\{ (x|y_1, ..., y_k) \middle| \forall S \subseteq [k] : (x + \sum_{i \in S} y_i) \in \tilde{X} \right\}$$

Using this definition, we can define a polynomial of degree $\leq d$ for subsets of $\mathbb{F}^n$:

Definition 3.2 (Polynomials in $\tilde{X}$)**.** *Let $d \in \mathbb{N}$ be an integer, and let $\tilde{X} \subseteq \mathbb{F}^n$. We say the degree of a function $f : \tilde{X} \to \mathbb{F}$ is d if d is the smallest integer such that f vanishes over all cubes of size $(d+1)$, i.e:*

$$\forall (x|y_1, ..., y_{d+1}) \in C_{d+1}(\tilde{X}) : D_{y_{d+1}} ... D_{y_1} p(x) = 0$$

A function over $\tilde{X}$ of degree $\leq d$ is also called a polynomial of degree $\leq d$*. We denote the set of polynomials of degree $\leq d$ over $\tilde{X}$ by $Poly_{\leq d}(\tilde{X} \to \mathbb{F})$.*

Note. *For $\tilde{X} = \mathbb{F}^n$, the definition above coincides with the local definition of polynomials.*

3.1 Lifting Polynomials

Our goal to achieve good properties for polynomials over $\tilde{X}$. To do so, we wish to connect the desired properties of polynomials defined over $\tilde{X}$, to properties known for polynomials over $\mathbb{F}^n$. Following such strategy raises a question: given a polynomial $p : \tilde{X} \to \mathbb{F}$, which polynomial over $\mathbb{F}^n$ should we consider to deduce properties of p? To find such a polynomial over $\mathbb{F}^n$, it would have been useful that all polynomials over $\tilde{X}$ actually "came from" polynomials over $\mathbb{F}^n$. More formally, it would have been useful that all polynomials $p : \tilde{X} \to \mathbb{F}$ would be equal to a restriction of some polynomial $P : \mathbb{F}^n \to \mathbb{F}$ of degree $\leq d$, to the set $\tilde{X}$. This would give us a "good candidate" (or candidates) to polynomials over $\mathbb{F}^n$, that using their known properties, we could achieve the properties we desire for polynomials over $\tilde{X}$.
Generally speaking, the existence of such polynomial $P : \mathbb{F}^n \to \mathbb{F}$ is not trivial by itself, and it mapy depend on the polynomial p and the set $\tilde{X}$. In this subsection, we discuss sets $\tilde{X} \subseteq \mathbb{F}^n$ that have this property for every polynomial $p : \tilde{X} \to \mathbb{F}$. Before formulating the notion above, we start by a simple remark:

Remark 3.3. *By the local criteria for $\mathbb{F}^n$, we have that a restriction of a polynomial of degree $\leq d$ over $\mathbb{F}^n$ to $\tilde{X}$ is a polynomial of degree $\leq d$ over $\tilde{X}$. Therefore, the other direction is true: every restriction of a polynomial over $\mathbb{F}^n$ to $\tilde{X}$ is a polynomial over $\tilde{X}$.*

Next, let us define subsets $\tilde{X} \subseteq \mathbb{F}^n$ that have the desired property, which we call *d-lift-enabler variety.*

Definition 3.4 (d-lift-enabler Subset)**.** *Let $\mathbb{F}$ be a field, and $n > 0$ be an integer. For an integer $d > 0$, we say a subset $\tilde{X} \subseteq \mathbb{F}^n$ is* d-lift-enabler *if for every $d' \leq d$, for every polynomial $p \in Poly_{d'}(\tilde{X} \to \mathbb{F})$ there exist a polynomial $\widehat{p} \in Poly_{d'}(\mathbb{F}^n \to \mathbb{F})$ such that $p|_{\tilde{X}} = \widehat{p}|_{\tilde{X}}$.*

Remark 3.5. *Using the local criterion of polynomials and the fact that that $C_{d+1}(\tilde{X}) \subseteq C_{d+1}(\mathbb{F}^n)$, one can see that for a polynomial $p : \tilde{X} \to \mathbb{F}$ with $\deg(p) = d$, every extension $P : \mathbb{F}^n \to \mathbb{F}$ with $p = P|_{\tilde{X}}$ holds the bound $\deg(\widehat{p}) \geq d$. The other direction is not true in the general case, but it is specifically promised when the variety is d-lift-enabler.*

This definition naturally raises the following definition:

Definition 3.6 (The Lift Operator)**.** *Let $d \in \mathbb{N}$ be an integer. Let $\tilde{X} \subseteq \mathbb{F}^n$ be a d-lift-enabler subset. We define* the d-lift operator *to be an operator $\widehat{\square} : Poly_{\leq d}(\tilde{X} \to \mathbb{F}) \to Poly_{\leq d}(\mathbb{F}^n \to \mathbb{F})$ the following way:*
Let $d' \leq d$. Given a polynomial $p : \tilde{X} \to \mathbb{F}$ of degree d', the operator $\widehat{\square}$ returns a polynomial $\widehat{p} : \mathbb{F}^n \to \mathbb{F}$ of degree d' such that $p = \widehat{p}|_{\tilde{X}}$. Note that we did not require the lift to be unique. Thus, in case there are multiple valid lifts for a polynomial $p \in Poly_{\leq d}(\tilde{X} \to \mathbb{F})$, the lift operator picks a single (consistent) one of them. Moreover, the lift always exists because the subset $\tilde{X}$ is d-lift-enabler. In addition, for a collection $\mathfrak{p} = (p_1, ..., p_c)$ of polynomials $p_i \in Poly_{\leq d}(\tilde{X} \to \mathbb{F})$, we denote $\widehat{\mathfrak{p}} \coloneqq (\widehat{p_1}, ..., \widehat{p_c})$

In the following subsections, we give example to two concrete sets $\tilde{X} \subseteq \mathbb{F}^n$ that are d-lift-enablers. Before doing so, we define an algebraic variety:

Definition 3.7 (Algebraic Variety)**.** *For a collection of functions $\mathfrak{F} \coloneqq \{F_1, ...F_c\}$ such that $F_i : \mathbb{F}^n \to \mathbb{F}$, we denote $Z(\mathfrak{F}) \coloneqq \{x \in \mathbb{F}^n | \forall i : F_i(x) = 0\}$.*
If the collection is a collection of polynomials, we call $Z(\mathfrak{F})$ an algebraic variety*, shorthand by* variety*.*
The degree of the variety is defined to be the maximal degree of polynomials in the collection that defines it. The rank of the variety is defined to be the rank of the collection that defines the variety (as a collection).

3.2 High Rank Varieities of High Minimal Degree

We now present a theorem proved in [KZ18, Corollary 1.10], that shows that high rank varieties are d-lift-enabler when the polynomials defining the variety are of degree $> d$:

Theorem 3.8. *Let $\mathbb{F}$ be a finite field, and let $\tilde{d}$, $\tilde{c} > 0$ representing parameters of a variety. Let $d < \tilde{d}$ a positive integer representing a degree of a polynomial which we wish to lift. There exists $\bar{r} = \bar{r}(\mathbb{F}, \tilde{d}, \tilde{c}) > 0$ such that for for all $n \in \mathbb{N}$, any variety $\tilde{X} = Z\left(\tilde{\mathcal{L}}\right) \subseteq \mathbb{F}^n$ for $\tilde{\mathcal{L}} = (L_1, ..., L_{\tilde{c}})$ such that rank $\left(\tilde{\mathcal{L}}\right) > \bar{r}$, degree $\deg(\tilde{\mathcal{L}}) = \tilde{d}$, with all defining polynomials of degree $\deg(L_i) > d$, it holds that $\tilde{X}$ is a d-lift-enabler subset.*

Remark 3.9. *Under the conditions stated above, it was proved in [KZ18] that the lift is in fact* unique*. Formally, if $p : \tilde{X} \to \mathbb{F}$ is a polynomial of degree $\leq d$, then there exists a* unique *polynomial $P : \mathbb{F}^n \to \mathbb{F}$ such that $P|_{\tilde{X}} \equiv p$.*

3.3 High Rank Varieties on a Large Field

In this subsection, we recall a theorem proved by [KZ19, Theorem 1.7] regarding high rank varieties that are defined on "large" fields. We note that the fields are large in respect of the degree d one wish to lift, but still does not depend on n.

Next we define a weakly polynomial, which generalizes our definition of a polynomial in $\tilde{X}$, that was used in [KZ19, Definition 1.1]:

Definition 3.10. *Let $\tilde{X} \subseteq \mathbb{F}^n$ be a set. We say a function $F : \tilde{X} \to \mathbb{F}$ is a* weakly polynomial of degree $\leq d$ *if for any affine subspace $L \subseteq \tilde{X}$, the restriction $F|_L$ is a polynomial of degree $\leq d$.*

Remark 3.11. *By the local criteria of a polynomial, it is easy to see that every $P \in Poly_{\leq d}(\tilde{X} \to \mathbb{F})$ is a weakly polynomial of degree $\leq d$.*

And now, we can present the lifting theorem for large fields, as proved in [KZ20, Theorem 2.17].

Theorem 3.12. *[KZ20, Theorem 2.17] Let $d, \tilde{d} \in \mathbb{N}$, and let $\mathbb{F}$ be a finite field such that $|\mathbb{F}| > d \cdot \tilde{d}$. There exists $r_{3.12} = r_{3.12}(\tilde{d}, d)$ such that for any variety $\tilde{X} \subseteq \mathbb{F}^n$ of maximal degree $\leq \tilde{d}$ and rank $\geq r_{3.12}$, have the following property: Every weakly polynomial function $p : \tilde{X} \to \mathbb{F}$ of degree $\leq d$ can be lifted to a polynomial function $P : \mathbb{F}^n \to \mathbb{F}$ of degree $\leq d$.*

Note. *Note that we stated the theorem above to finite fields, but it is also valid for infinite algebraically closed fields.*

The theorem above implies the following corollary:

Corollary 3.13. *Let $d, \tilde{d} \in \mathbb{N}$, and let $\mathbb{F}$ be a finite field such that $|\mathbb{F}| > d \cdot \tilde{d}$. There exists $r_{3.12} = r_{3.12}(\tilde{d}, d)$ such that for any variety $\tilde{X} \subseteq \mathbb{F}^n$ of degree $\tilde{d}$ and rank $\geq r_{3.12}$ is a d-lift-enabler.*

4 Relative Rank-Bias Property

In this section, we generalize the relation between rank and bias that is known for $\mathbb{F}^n$ also for $\tilde{X} \subseteq \mathbb{F}^n$. Specifically, in Lemma 2.12, it was shown that high rank factors have low bias in $\mathbb{F}^n$. We wish to define an alternative definition of rank for $\tilde{X} \subseteq \mathbb{F}^n$, called *$\tilde{X}$-relative rank*, such that high $\tilde{X}$-relative rank implies low bias in $\tilde{X}$. This type of relation (and definition) was shown previously to a few sets; in [LZ21, Theorem 1.8] for sets $\tilde{X} = Z(\mathcal{Q})$ where $\mathcal{Q}$ is a collection of polynomials of high rank; and in [GK22b, Theorem 1.4] for sets $\tilde{X} = S^n$ for $S \subset \mathbb{F}$.

To understand this notion, we first introduce a simple example that demonstrates the need for a different definition of rank to achieve equidistribution properties in subsets of $\mathbb{F}^n$.

Example 4.1. *Let $\tilde{X} = \{x \in \mathbb{F}^n | x_1 = 0\}$. Define $P : \mathbb{F}^n \to \mathbb{F}$ by $P(x) \coloneqq x_1$. In $\mathbb{F}^n$, P has rank ∞ as it can not be decomposed polynomials of degree < 1 (constants). Additionally, it is perfectly equidistributed. This is the simplest example of the rank-bias relation in $\mathbb{F}^n$.*
However, when restricting P to $\tilde{X}$, we get $P|_{\tilde{X}} \equiv 0$. As 0 is a constant function, it is the least *equidistributed possible in $\tilde{X}$. Therefore, we see that the way we defined rank in $\mathbb{F}^n$ does not imply the desired equidistribution in $\tilde{X}$: we found a polynomial with high rank (infinity) that has a very high bias* in $\tilde{X}$ *(the maximal).*

The reason the definition of rank in $\mathbb{F}^n$ fails to capture equidistribution even on subsets that are really similar to $\mathbb{F}^n$ (isomorphic to $\mathbb{F}^k$), is because of the following reason: Even though our polynomial P does *not* have a decomposition to a few lower-degree polynomials by itself, there *exists* a *$\tilde{X}$-equivalent* polynomial that has such structured decomposition. Here, by $\tilde{X}$-equivalent we mean a polynomial in $\mathbb{F}^n$ that is bounded by the same degree bound, and is equal to P in $\tilde{X}$. In the example described above, this equivalent polynomial

is the constant function 0, and its decomposition is the trivial one (any function decomposes a constant function). An alternative perspective which we use throughout this paper to $\tilde{X}$-equivalence is that both polynomials are equal up to a *valid $\tilde{X}$-remainder*: a bounded degree polynomial that is $\equiv 0$ in $\tilde{X}$.

Generally speaking, high $\tilde{X}$-relative rank may not imply low bias in $\tilde{X}$. Therefore, this structure-randomness relation is not true for a general subset $\tilde{X} \subseteq \mathbb{F}^n$, but is a *property* of the subset $\tilde{X}$. Thus, we say that a subset has the *relative rank-bias property* if this relation holds, i.e. if high $\tilde{X}$-relative rank implies equidistribution in $\tilde{X}$.

Let us now formally define our definition for relative rank, inspired by the two different definitions of relative rank presented in [LZ21, Definition 1.6] and in [GK22b, Definition 1.3]:

Definition 4.2 (Relative Rank of a Polynomial)**.** *Let $\tilde{X} \subseteq \mathbb{F}^n$ and let $d \in \mathbb{N}$. For an integer $d \in \mathbb{N}$ and a polynomial $P : \mathbb{F}^n \to \mathbb{F}$, we define its d-*relative rank *in respect of $\tilde{X}$ as:*

$$rank_{d,\tilde{X}}(P) \coloneqq \min\left\{rank_d\left(P - \overline{P}\right) \middle| \overline{P} \in Poly_{\leq \deg(P)}(\mathbb{F}^n \to \mathbb{F}), \overline{P}|_{\tilde{X}} \equiv 0\right\}$$

For a polynomial P of degree $\deg(P) = d$ we denote $rank_{\tilde{X}}(P) \coloneqq rank_{d,\tilde{X}}(P)$.

Definition 4.3 ($\tilde{X}$-equivalent and $\tilde{X}$-remainder)**.** *Moreover, we say a polynomial is $\tilde{X}$-*equivalent *to P if its restriction to $\tilde{X}$ is $\equiv P|_{\tilde{X}}$. We say it is* valid $\tilde{X}$-equivalent to P *if it is $\tilde{X}$-equivalent to P and its of the* same *degree of P.*
*Similarly, we say a polynomial is $\tilde{X}$-*remainder *of P if its restriction to $\tilde{X}$ is $\equiv 0$. We say it is* valid $\tilde{X}$-remainder *if it is $\tilde{X}$-remainder of P and its of degree* smaller or equal *of the degree of P.*
We typically denote such polynomial as $\overline{P}$.

In other words, the d-relative rank of a polynomial P is the smallest d-rank of all valid $\tilde{X}$-equivalents of P.

Note 4.4. *Note that [GK22b, Definition 1.3] defines rank in a substantially different way than our definition, and consequentially our results will not apply to the sets they presented. One of the main differences in the definition of rank occurs for $d = 1$. In the definition we use for rank, the rank of every (non-constant) degree-1 polynomial is ∞, where in the definition used in [GK22b] it is a finite number (which is possibly very small). This difference is crucial, as for example, it makes regularization according to their definition not-trivially possible, where it is known to be possible when rank is defined by the definition we use (Lemma 2.21).*

Definition 4.5 (Relative Rank of a Factor)**.** *Let $\tilde{X} \subseteq \mathbb{F}^n$. Let $\mathcal{P}$ be a set of polynomials $\mathcal{P} = \{P_1, ..., P_c\}$. The rank of the polynomial set $\mathcal{P}$ relative to the subset $\tilde{X}$ is defined as:*

$$rank_{\tilde{X}}(\mathcal{P}) \coloneqq \min\left\{ rank_{d,\tilde{X}}\left(\sum_{i=1}^{c} \lambda_i P_i\right) \middle| 0 \neq \vec{\lambda} \in \mathbb{F}^c, d = \max_{i\in[c]} \deg(\lambda_i P_i) \right\}$$

For a factor $\mathcal{B}$ defined by a collection of polynomials, we define its relative rank relative to $\tilde{X}$ to be the relative rank of the collection of polynomials defining it, relative to the set $\tilde{X}$.
For a non-decreasing function $r : \mathbb{N} \to \mathbb{N}$, a factor $\mathcal{B}$ is called r-$\tilde{X}$-regular if its relative rank in respect to $\tilde{X}$ is at least $r(|\mathcal{B}|)$.

4.1 Relative Rank-Bias Property

Definition 4.6 (Relative Rank-Bias property)**.** *Let $\mathbb{F}$ be a finite field, and let $d \in \mathbb{N}$ be an integer. Let $\tilde{r} : \mathbb{R}^+ \to \mathbb{N}$ be a function that represents the rank-bias relation for a fixed $d, \mathbb{F}$.*
We say a set $\tilde{X} \subseteq \mathbb{F}^n$ has the $(\tilde{r}, \mathbb{F}, d)$-relative rank-bias property *if for every $\epsilon > 0$, for every polynomial P of degree $\leq d$ with $rank_{\tilde{X}}(P) \geq \tilde{r}(\epsilon)$ we have:*

$$bias_{\tilde{X}}(P) < \epsilon$$

As an immediate corollary of Lemma 2.12 that shows that high rank implies low bias, we have that $\tilde{X} = \mathbb{F}^n$ has the relative rank-bias property.

Corollary 4.7 ($\mathbb{F}^n$ has the relative rank-bias property)**.** *For every finite field $\mathbb{F}$ and $d \in \mathbb{N}$, let $\tilde{r} : \mathbb{R}^+ \to \mathbb{N}$ defined as $\tilde{r}(\epsilon) \coloneqq r_{2.12}(\mathbb{F}, d, \epsilon)$. Then, we have that the set $\tilde{X} = \mathbb{F}^n$ has the $(\tilde{r}, \mathbb{F}, d)$-relative rank-bias property.*

Proof. This is a simple usage of Lemma 2.12: Note that when $\tilde{X} = \mathbb{F}^n$, we have that $rank_{\tilde{X}}(P) = rank(P)$. Now, if P is a polynomial of degree $\leq d$ and $rank_{\tilde{X}}(P) = rank(P) \geq \tilde{r}(\epsilon) = \tilde{r}_{2.12}(\mathbb{F}, d, \epsilon)$, then:

$$bias_{x \in \mathbb{F}^n}(P(x)) < \epsilon$$

□

4.2 Limited-Relative Rank-Bias Property

Sometimes, however, we can not request $\tilde{X}$ to be such that high relative rank implies low bias for every $\epsilon > 0$, but only for $\epsilon' \geq \epsilon$ for some constant $\epsilon > 0$. This leads to defining the *limited relative rank-bias property*, which will be used to discuss such sets $\tilde{X} \subseteq \mathbb{F}^n$.
As we will later see, this definition raises naturally where $\tilde{X}$ is a high rank variety, in which for the relative rank-bias property to hold for some $\epsilon > 0$, the rank of the variety should be greater than a value that is dependent of ϵ. Thus, to have the relative rank-bias property for a high rank variety but without requiring an infinitely large rank, we must limit the relative rank-bias property for $\epsilon' \geq \epsilon$ We formulate the definition of this property as follows:

Definition 4.8 (Limited Relative Rank-bias property)**.** *Let $\mathbb{F}$ be a finite field, let $d \in \mathbb{N}$ be an integer, and let $\epsilon > 0$ be a constant. Let $\tilde{r} : [\epsilon, \infty] \to \mathbb{N}$ be a function that represents the limited-relative-rank-bias relation.*
We say a set $\tilde{X} \subseteq \mathbb{F}^n$ has the $(\tilde{r}, \mathbb{F}, d, \epsilon)$-limited-relative-rank-bias property *if for every $\epsilon' \geq \epsilon$, for every polynomial P of degree $\leq d$ with $rank_{\tilde{X}}(P) \geq \tilde{r}(\epsilon')$ we have:*

$$bias_{\tilde{X}}(P) < \epsilon'$$

As a convention, we denote by $\tilde{\epsilon}$ the ϵ such that the limited-relative-rank-bias property holds for $\tilde{X}$.

4.2.1 High Rank Varieities

In this subsection, we are discussing specifically $\tilde{X} \subseteq \mathbb{F}^n$ that are in the form $\tilde{X} = Z(\mathcal{Q})$ for a set of polynomials $\mathcal{Q}$ that form a high rank factor. Let

us present some known results of the relative rank-bias relation for high rank varieites: In the scenario when we are working relative to $\tilde{X}$, the equivalent for Theorem 2.12 is also known when we assume $d < char(\mathbb{F})$, as shown in [LZ21, Theorem 1.8]:

Theorem 4.9 (High relative rank implies low bias in high rank varieties)**.** *Let $\mathbb{F}$ be a finite field and let $0 \leq d < char(\mathbb{F})$. Let $\epsilon > 0$ be a constant, and let $\tilde{c} \in \mathbb{N}$. There exist $\bar{r}^{4.9} = \bar{r}^{4.9}(\mathbb{F}, d, \tilde{c}, \epsilon)$ and $r^{4.9} = r^{4.9}(\mathbb{F}, d, \epsilon)$ such that the following holds:*
Let $\tilde{\mathcal{L}} = (L_1, ..., L_{\tilde{c}})$ be a collection of polynomials of degrees $\leq d$ with rank $\left(\tilde{\mathcal{L}}\right) \geq \bar{r}^{4.9}$ and let P be a polynomial of degree $\leq d$.
Then, if $rank_{\tilde{\mathcal{L}}}(P) \geq r^{4.9}$, we have:

$$bias_{\tilde{\mathcal{L}}}(P) < \epsilon$$

Note. *Note that the original statements in [LZ21] are stated for a different definition of rank, noted as* schmidt rank*. In the appendix 8 we compare the two different definitions, and show that our definition of rank is comprehensive enough in a sense that a polynomial with high rank also has high schmidt rank. Additionally, we show that for a given $r \in \mathbb{N}$, the lower bound of rank required for a polynomial to be of schmidt rank $\geq r$, is only $c \cdot r$ for some constant $c \in \mathbb{N}$.*

Remark 4.10. *Note that in the original statement of theorem 4.9 as stated in [LZ21, Theorem 1.8], there are good bounds on the rank needed for $\tilde{\mathcal{L}}$ and P for the theorem to hold.*
Specifically, there exist constants $A(d), B(d)$ such that for an error $\epsilon = |\mathbb{F}^{-s}|$, if $\bar{r}^{4.9} = A(\tilde{c} + s)^B$ and $r^{4.9} = A(1 + s)^B$, then we have:

$$bias_{Z\left(\tilde{\mathcal{L}}\right)}(P) < |\mathbb{F}|^{-s}$$

In our proof, it is enough that the bounds on r and $\bar{r}$ are independent of n, thus we omit the exact bounds stated above and use the statement as stated in Theroem 4.9.

Remark 4.11. *Note that both $r^{4.9}(\mathbb{F}, d, \epsilon)$ and $\bar{r}^{4.9}(\mathbb{F}, d, \tilde{c}, \epsilon)$ are decreasing when ϵ is increasing. This means for example, that for all $\epsilon' \geq \epsilon$, a variety that satisfies the theorem's rank condition for ϵ also satisfies the theorem's rank condition for ϵ'. Therefore, a polynomial with rank $\geq r^{4.9}(\mathbb{F}, d, \epsilon')$ will have a bias $< \epsilon'$.*

As a corollary of Theorem 4.9 and Corollary 8.7, we have that high rank varieties has the limited-relative-rank-bias property. Formally, we have:

Corollary 4.12 (High Rank Varieties Have the Limited-Relative Rank-Bias Property)**.** *Let $\mathbb{F}$ be a finite field, and let $\tilde{d} \in \mathbb{N}$ such that $0 < \tilde{d} < |\mathbb{F}|$. Let $\tilde{\epsilon} > 0$ be a constant which represents the desired relative rank-bias limit. There exists $\tilde{r}_{4.12} : [\tilde{\epsilon}, \infty] \to \mathbb{N}$ with $\tilde{r}_{4.12} \coloneqq \tilde{r}_{4.12}(\mathbb{F}, \tilde{d})$ such that the following holds:*
Let $\tilde{c} \in \mathbb{N}$ be an integer. There exists $\bar{r}_{4.12} \coloneqq \bar{r}_{4.12}(\mathbb{F}, \tilde{d}, \tilde{c}, \tilde{\epsilon})$ such that for every $\tilde{\mathcal{L}} = (L_1, ..., L_{\tilde{c}})$ polynomial factor of degree $\leq \tilde{d}$ with rank $\left(\tilde{\mathcal{L}}\right) \geq \bar{r}$, we have:
The variety $\tilde{X} = Z\left(\tilde{\mathcal{L}}\right)$ has the $(\tilde{r}_{4.12}, \mathbb{F}, \tilde{d}, \tilde{\epsilon})$-limited-relative-rank-bias property.

Proof. Let $\mathbb{F}$ be a finite field, and let $\tilde{d} \in \mathbb{N}$ such that $0 < \tilde{d} < |\mathbb{F}|$. Let $\tilde{\epsilon} > 0$. We choose:

$$\tilde{r}_{4.12}(\epsilon) \coloneqq r_{4.9}(\mathbb{F}, \tilde{d}, \epsilon)$$

Note that for every ϵ in its domain, $\tilde{r}_{4.12}$ does not depend on $\tilde{\epsilon}$. Let $\tilde{c} \in \mathbb{N}$. Now, we choose:

$$\bar{r}_{4.12}(\mathbb{F}, \tilde{d}, \tilde{c}, \tilde{\epsilon}) \coloneqq \bar{r}_{4.9}(\mathbb{F}, \tilde{d}, \tilde{c}, \tilde{\epsilon})$$

Using Theorem 4.9 that shows high rank implies low bias in $\tilde{X}$, and the assumption that $\epsilon \geq \tilde{\epsilon}$ (specifically Remark 4.11) concludes the proof. □

5 Regularization Relative to $\tilde{X}$

In this section, we generalize the definitions and statements regarding factors and regularization in $\mathbb{F}^n$, to their corresponding definitions and statements to relative rank in respect of $\tilde{X} \subseteq \mathbb{F}^n$.
Note that in oppose to the previous chapter that we discussed a general U and $A \subseteq U$, in this chapter we discuss only $U = A = \mathbb{F}^n$. This is done for clearance and to avoid defining definitions we will not use in our main proof.

Definition 5.1 (Measurable Relative to $\tilde{X}$)**.** *Let $\mathfrak{F} = \{F_1, ..., F_c\}$ be a set of functions $F_i : \mathbb{F}^n \to \mathbb{F}$. We say a function $G : \mathbb{F}^n \to \mathbb{F}$ is* measurable in respect of $\mathfrak{F}$ relative to $\tilde{X}$*, or* $\tilde{X}$-relative $\mathfrak{F}$-measurable*, if there exists a function $\overline{G} : \mathbb{F}^n \to \mathbb{F}$ with $\overline{G}|_{\tilde{X}} \equiv 0$ and a function $\Gamma : \mathbb{F}^c \to \mathbb{F}$ such that:*

$$\forall a \in A : G(a) = \Gamma(F_1(a), ..., F_c(a)) + \overline{G}(a)$$

And:

$$\deg(G - \overline{G}) \leq \deg(G)$$

We sometimes refer to Γ as the $\tilde{X}$-relative measurement function*.*

Note. *Note that if $\deg(G - \overline{G}) \leq \deg(G)$ as discussed above, then the same bound also bounds the degree of the remainder, i.e. $\deg(\overline{G}) \leq \deg(G)$. Therefore $\overline{G}$ is a valid $\tilde{X}$-remainder of G. Moreover, this requirement is equivalent to the definition above, as if $\deg(\overline{G}) \leq \deg(G)$, then we also have $\deg(G - \overline{G}) \leq \deg(G)$.*

Note. *Also note that without the bound on the degree of the remainder, being measurable relative to $\tilde{X}$ is in fact equivalent for being a measurable in $A = \tilde{X}$. This is true because under these conditions, the remainder $\overline{G}$ has no constraints but $\overline{G}|_{\tilde{X}} \equiv 0$, thus the condition left on the measurement is just being a measurement to G in $\tilde{X}$.*

Remark 5.2. *If G is a function that it is $\mathfrak{F}$-measurable relative to $\tilde{X}$, then every value of G can be determined by the values of $F_1, ..., F_c$ up to a remainder $\overline{G}$ of degree $\leq d$. Thus, perhaps we do not know that the function G is constant inside every atom of $\mathfrak{F}$ as in a regular semantic refinement, but we do know that there exists a function $(G - \overline{G})$ that equals to G on $\tilde{X}$, is constant on every atom of $\mathfrak{F}$ and it is a function with a bounded degree i.e. $\deg(G - \overline{G}) \leq \deg(G)$.*

Next, we present a new type of refinement, which is a relaxation of semantic refinement. This relaxation will allow us to discuss the corresponding claim of the polynomial regularity lemma (Lemma 2.21) for relative rank (instead of rank).

Definition 5.3 (Semantic Refinement Relative to $\tilde{X}$)**.** *Let $\mathcal{B}$ and $\mathcal{B}'$ be polynomial factors on $\mathbb{F}^n$, defined by sets of polynomials $\mathcal{P}, \mathcal{P}'$ respectively, and let $d \in \mathbb{N}$. We say a factor $\mathcal{B}'$ is a* semantic refinement relative to $\tilde{X}$ *of the factor $\mathcal{B}$, or* $\tilde{X}$-relative semantic refinement*, if the following holds: Every function $F : \mathbb{F}^n \to \mathbb{F}$ that is $\mathcal{P}$-measurable relative to $\tilde{X}$, is also $\mathcal{P}'$-measurable relative to $\tilde{X}$. If the definition above holds, we denote $\mathcal{B}' \succeq^{\tilde{X}}_{sem} \mathcal{B}$.*

Note. *It is easy to see that this relation is transitive, i.e. if $\mathcal{B}' \succeq^{\tilde{X}}_{sem} \mathcal{B}$ and $\mathcal{B}'' \succeq^{\tilde{X}}_{sem} \mathcal{B}'$, then $\mathcal{B}'' \succeq^{\tilde{X}}_{sem} \mathcal{B}$.*

Remark 5.4. *In $\tilde{X}$, semantic refinements relative to $\tilde{X}$ behave the same as regular semantic refinements in the perspective of being measurable: every function that is $\mathcal{P}$-measurable in $\tilde{X}$ is also $\mathcal{P}'$-measurable in $\tilde{X}$. However, the two definitions behave differently in the perspective of being measurable in $\mathbb{F}^n$. Specifically, in relative semantic refinements, if G is a $\mathcal{P}$-measurable function it is not necessarily $\mathcal{P}'$-measurable. However, it is measurable up to a remainder $\overline{G}$ of degree $\leq \deg(G)$ such that $\overline{G}|_{\tilde{X}} \equiv 0$.*

Corollary 5.5. *If $\mathcal{B}' \succeq^{\tilde{X}}_{sem} \mathcal{B}$, then in $\tilde{X}$ it is a regular semantic refinement, i.e. $\mathcal{B}' \succeq_{sem|\tilde{X}} \mathcal{B}$.*

Next, we present a new regularization process that allows us to increase the *relative* rank of a factor without increasing the size of the factor too much (independent of n). This regularization process generalizes the regularization process in $\mathbb{F}^n$, which was first presented by [GT07a, 2.3]. We call this type of regularization process a *relative-regularization process* relative to $\tilde{X}$, shorthand by $\tilde{X}$-regularization For a specific function r, we will sometimes call applying this lemma a *r-$\tilde{X}$-regularization*. Note that to allow such a relative-regularization process to hold, we must use the relaxed definition of semantic refinement that is presented above.

Theorem 5.6. *Let $r : \mathbb{N} \to \mathbb{N}$ be a non-decreasing function and let $d \in \mathbb{N}$. There exists $C^{5.6}_{r,d} : \mathbb{N} \to \mathbb{N}$ such that the following holds: Let $\mathcal{B}$ be a factor defined by polynomials $\mathcal{P} = (P_1, ..., P_c)$ where for all $i \in [c]$: $P_i : \mathbb{F}^n \to \mathbb{F}$ and $\deg(P_i) \leq d$. Then, there is an r-$\tilde{X}$-regular factor $\mathcal{B}'$ defined by polynomials $\mathcal{P}' = (P'_1, ..., P'_{c'})$ where for all $i \in [c]$: $P'_i : \mathbb{F}^n \to \mathbb{F}$ and $\deg(P'^{]}_i) \leq d$ such that $\mathcal{B}' \succeq^{\tilde{X}}_{sem} \mathcal{B}$ and $c' \leq C^{5.6}_{r,d}(c)$.*
Moreover, if $\mathcal{B} \succeq_{syn} \bar{\mathcal{B}}$ for some polynomial factor $\bar{\mathcal{B}}$ with relative rank of at least $r(c') + c' + 1$ and rank of at least $r_{2.19}(\mathbb{F}, d, c') + c' + 1$, then we can require that $\mathcal{B}' \succeq_{syn} \bar{\mathcal{B}}$.

Proof. We follow the lines of the proof given by [HHL19][Lemma 7.29], but here, we wish to increase the *relative* rank of the factor instead of its rank. We present an iterative process, which will eventually lead us to a factor of size c' with relative rank higher than $r(c')$, that is a semantic refinement relative to $\tilde{X}$. Let $d \in \mathbb{N}$, and let $\mathcal{B}$ be a polynomial factor defined by $\mathcal{P} = (P_1, ..., P_c)$ such that $P_i : \mathbb{F}^n \to \mathbb{F}$ of degree $\leq d$. Define $M(\mathcal{B}) \coloneqq (M_d, ..., M_1) \in \mathbb{N}^d$, where M_i denotes the number of polynomials in $\mathcal{P}$ that have degree exactly i. Thus, $\sum_{i=1}^{d} M_i = c$. We define the lexicographical order on $\mathbb{N}^d$ where $M > M'$ if and only if $M_i > M'_i$ for some $1 \leq i \leq d$, and $M_j = M'_j$ for all $j > i$. This proof will be by transfinite induction on M under the lexicographical order. Next we

describe a step of the regularization process.
Let $\mathcal{B}$ be a polynomial factor defined by $\mathcal{P} = (P_1, ..., P_c)$. Note that this is an abuse of notations: the factor $\mathcal{B}$ and the set $\mathcal{P}$ refer to the original factor in the first step, and also to the current factor in the middle of the relative-regularization process. If $\mathcal{B}$ is r-$\tilde{X}$-regular, then we are done. Otherwise, we change $\mathcal{B}$ as follows: First, we denote $r_{2.19}^{\mathbb{F},d}(c) \coloneqq r_{2.19}(\mathbb{F}, d, c)$, and we $r_{2.19}$-regularize $\mathcal{P}$ using lemma 2.21 to get a set of polynomials $\mathcal{P}_1 = (P_1^1, ..., P_{c_1}^1)$ of degree $\leq d$, which defines a factor $\mathcal{B}_1$ and has a rank $\geq r_{2.19}^{\mathbb{F},d}(c_1)$. Note that $M(\mathcal{B}_1) \leq M(\mathcal{B})$. Then, again, if somehow $\mathcal{B}_1$ is now r-$\tilde{X}$-regular, we are done. Otherwise, by definition, there exists some linear combination of the polynomials in $\mathcal{P}_1$ that has $d^\star$-relative rank less than $r(c_1)$, where $d^\star$ is the maximal degree that participates in the linear combination. Let $\vec{P}(x) = \sum_{i=0}^{c_1} \lambda_i P_i^1(x)$ where $\vec{0} \neq \vec{\lambda} \in \mathbb{F}^{c_1}$, be the linear combination with $rank_{d^\star,\tilde{X}}\left(\vec{P}\right) \leq r(c_1)$ where $d^\star \coloneqq \max_{i\in[c_1]} \deg(\lambda_i P_i^1)$. By definition of relative rank, there exists $\overline{P} \in Poly_{\leq \deg(\vec{P})}(\mathbb{F}^n \to \mathbb{F})$ with $\overline{P}|_{\tilde{X}} \equiv 0$ such that $rank_{d^\star}\left(\vec{P} - \overline{P}\right) \leq r(c_1)$. Note that $\deg(\overline{P}) \leq d^\star$. By definition of $d^\star$-rank, we have that we can decompose $\vec{P} - \overline{P}$ as a function of $r(c_1)$ polynomials of degree $\leq d^\star - 1$. In other words, there exist a measurement function $\vec{\Gamma} : \mathbb{F}^{r(c_1)} \to \mathbb{F}$ and polynomials $Q_1, ..., Q_{r(c_1)}$ with $\deg(Q_i) \leq d^\star - 1$ such that:

$$\forall a \in \mathbb{F}^n : \vec{P}(a) - \overline{P}(a) = \vec{\Gamma}\left(Q_1(a), ..., Q_{r(c_1)}(a)\right)$$

Now, let $\mathcal{P}^\star \subseteq \mathcal{P}_1$ be the set of all such maximal-degree polynomials, and let $i^\star$ be chosen such that $P_{i^\star}^1 \in \mathcal{P}^\star$. Note that the set $\mathcal{P}^\star$ is non empty, as by definition, $d^\star$ is the maximal degree of polynomial in the expression $\sum_{i=1}^{c_1} \lambda_i P_i^1$ such that $\lambda_i \neq 0$.
For the next step, define the polynomial factor $\mathcal{B}_2$ be the polynomial factor defined by the set:

$$\mathcal{P}_2 \coloneqq \mathcal{P}_1 \setminus \left\{P_{i^\star}^1\right\} \cup \left\{Q_1, ..., Q_{r(c_1)}\right\}$$

Finally, the factor $\mathcal{B}_2$ will be the factor returned from the relative-regularization step.
It is easy to see that if the process above halts, we get a r-$\tilde{X}$-regular factor. Now, we prove the first part of the lemma by showing the following claims:

Claim 5.7. *The factor generated from the regularization above is of bounded size: a bound that may depend on r, d, c, but does not depend on n. Formally, we claim that there exists $C_{r,d}^{5.6} : \mathbb{N} \to \mathbb{N}$ such that we have $c' \leq C_{r,d}^{5.6}(c)$.*

Proof. It is enough to prove the following:

1. In each step, the amount of polynomials there are in $\mathcal{P}_1, \mathcal{P}_2$ are bounded by a bound that depend only on r, d, c (independent of n).

2. The number of steps of the relative-regularization process is also bounded by a bound that depends only on r, d, c (independent of n).

The combination of these two will obtain the desired bound of the amount of polynomials in the last-step regularized factor, which is $C_{r,d}^{5.6}(c)$. Note that the

bound on the last-step relative-regularized factor in not simply the multiplication of the two bounds, but a recursively-substitution of the bound in 1, a bounded amount of times (bounded by the bound in 2).
For 1, we first notice that the number of polynomials in the regular regularization process is bounded, specifically we have $\left|\mathcal{P}^1\right| = c_1 \leq C^{2.21}_{r_{2.19},d}(c)$. Moreover, the polynomial factor $\mathcal{B}_2$ is generated by adding at most $r(c_1)$ polynomials to the factor, and thus we have $|\mathcal{P}_2| \leq c_1 + r(c_1)$ which is also bounded by substituting the bound on c_1.
For 2, we use the transfinite induction on M we mentioned earlier to show that the process must halt after a bounded number of steps. Formally, we show that there exist M' which depends only on $M(\mathcal{B})$ such that $M(\mathcal{B}_2) \leq M' < M(\mathcal{B})$. This will bound the number of steps by a value that depend only on $M(\mathcal{B})$, which depends only on r, d, c. To do so, we first notice that the regular regularization does not increase the value of M, i.e. $M(\mathcal{B}_1) \leq M(\mathcal{B})$. Thus, we can focus on the second part of the relative-regularization. In this part, we replace a single degree $d^\star$ polynomial by at most $r(c_1)$ polynomials of degree $\leq d^\star - 1$. Therefore, by choosing $M' \coloneqq (M_d, ..., M_{d^\star+1}, M_{d^\star} - 1, M_{d^\star-1} + r(c_1), ..., M_1 + r(c_1))$ we get that $M(\mathcal{B}_2) \leq M' < M(\mathcal{B}_1) \leq M(\mathcal{B})$, which concludes 2. □

Claim 5.8. *The factor generated from the regularization above is a $\tilde{X}$-relative semantic refinement of the original factor, i.e $\mathcal{B}' \succeq^{\tilde{X}}_{sem} \mathcal{B}$.*

Proof. It is enough to show that in each step, the factors generated by the relative-regularization process are semantic refinements relative to $\tilde{X}$ of the previous step's factor. Specifically, we show $\mathcal{B}_2 \succeq^{\tilde{X}}_{\text{sem}} \mathcal{B}_1 \succeq^{\tilde{X}}_{\text{sem}} \mathcal{B}$ and the claim will follow from transitivity of relative semantic refinements.
We start by proving $\mathcal{B}_1 \succeq^{\tilde{X}}_{\text{sem}} \mathcal{B}$. Let $F : \mathbb{F}^n \to \mathbb{F}$ be a function that is $\mathcal{P}$-measurable relative to $\tilde{X}$. We denote $d_F \coloneqq \deg(F)$. By definition, there exists $\Gamma : \mathbb{F}^c \to \mathbb{F}$, $\overline{F} : \mathbb{F}^n \to \mathbb{F}$ where $\deg(\overline{F}), \deg(F - \overline{F}) \leq d_F$ and $\overline{F}|_{\tilde{X}} \equiv 0$, such that:

$$\forall a \in \mathbb{F}^n : F(a) = \Gamma(P_1(a), ..., P_c(a)) + \overline{F}(a)$$

Clearly, the function $\Gamma(P_1(a), ..., P_c(a))$ is $\mathcal{P}$-measurable in $\mathbb{F}^n$, and because we have $\mathcal{B} \succeq_{\text{sem}} \mathcal{B}_1$, it is also $\mathcal{P}_1$-measurable in $\mathbb{F}^n$. Thus there exists $\Gamma_1 : \mathbb{F}^{c_1} \to \mathbb{F}$ such that:

$$\forall a \in \mathbb{F}^n : F(a) = \Gamma_1(P^1_1(a), ..., P^1_{c_1}(a)) + \overline{F}(a)$$

And therefore we have $\mathcal{B}_1 \succeq^{\tilde{X}}_{\text{sem}} \mathcal{B}$.
Now, we prove $\mathcal{B}_2 \succeq^{\tilde{X}}_{\text{sem}} \mathcal{B}_1$. Let $F : \mathbb{F}^n \to \mathbb{F}$ be a function that is $\mathcal{P}_1$-measurable relative to $\tilde{X}$. Again, we denote $d_F \coloneqq \deg(F)$, and by definition there exists $\Gamma_1 : \mathbb{F}^c \to \mathbb{F}$, $\overline{F_1} : \mathbb{F}^n \to \mathbb{F}$ where $\deg(F - \overline{F_1}) \leq d_F$ and $\overline{F_1}|_{\tilde{X}} \equiv 0$, such that:

$$\forall a \in \mathbb{F}^n : F(a) = \Gamma_1(P^1_1(a), ..., P^1_{c_1}(a)) + \overline{F}_1(a) \tag{1}$$

Note that we also have $\deg(\overline{F_1}) \leq d_F$. We will refer this equation, and its simplifications we do throughout the proof, as *the $\mathcal{P}_1$-decomposition of F*.
We wish to show that there exists $\Gamma_2 : \mathbb{F}^{c_2} \to \mathbb{F}$ and $\overline{F}_2 : \mathbb{F}^n \to \mathbb{F}$ where $\deg(F - \overline{F}_2) \leq d_F$ and $\overline{F}_2|_{\tilde{X}} \equiv 0$, such that:

$$\forall a \in \mathbb{F}^n : F(a) = \Gamma_2\left(P^1_1(a), ...P^1_{i^\star-1}(a), P^1_{i^\star+1}(a), ..., P^1_c(a), Q_1(a), ..., Q_{r(c_1)}(a)\right) + \overline{F}_2(a)$$

We will do so using the $\mathcal{P}_1$-decomposition of F. Note that showing $\deg(F-\overline{F}) \leq d_F$ is equivalent of showing $\deg(\overline{F}_2) \leq d_F$.
First, by the way we built $\mathcal{P}_2$, using the same notations in the regularization step, we have:

$$\forall a \in \mathbb{F}^n : P^1_{i^\star}(a) = \vec{\Gamma}\left(Q_1(a), ..., Q_{r(c_1)}(a)\right) + \overline{P}(a) - \sum_{i \neq i^\star} P^1_i(a)$$

Next, we substitute the value of $P^1_{i^\star}$ in the $\mathcal{P}_1$-decomposition of F (1), and get another decomposition of F that does not depend on $P^1_{i^\star}$. Specifically we have:

$$\forall a \in \mathbb{F}^n : F(a) = \Gamma_1 \left(P^1_1(a), ..., \left(\vec{\Gamma}\left(Q_1(a), ..., Q_{r(c_1)}(a)\right) + \overline{P}(a) - \sum_{i \neq i^\star} P^1_i(a) \right), ..., P^1_{c_1}(a)) \right) \tag{2}$$

$$+ \overline{F}_1(a) \tag{3}$$

We wish to use the equation above to show that F is $\mathcal{P}_2$-measurable relative to $\tilde{X}$. However, in order to show that the equation above is in the desired structure that proves that F is $\mathcal{P}_2$-measurable, the expression inside Γ_1 must not depend on $\overline{P}$ because $\overline{P} \notin \mathcal{P}_2$. Note that this is enough as the rest of the polynomials in the expression above are in $\mathcal{P}_2$, and therefore without $\overline{P}$ the expression is $\mathcal{P}_2$-measurable.
To do so, we start by simplifying some of the notations. We denote:

$$\vec{P}_2(a) \coloneqq \vec{\Gamma}\left(Q_1(a), ..., Q_{r(c_1)}(a)\right) - \sum_{i \neq i^\star} P^1_i(a)$$

This is the part of the sum that decomposes $P^1_{i^\star}(a)$ that is $\mathcal{P}_2$-measurable, thus the following equality applies:

$$P^1_{i^\star}(a) = \vec{P}_2(a) + \overline{P}(a)$$

where $\deg(\vec{P}_2), \deg(\overline{P}) \leq d^\star$. Using this notation, we write the $\mathcal{P}_1$-decomposition of F (2), and get:

$$\forall a \in \mathbb{F}^n : F(a) = \Gamma_1 \left(P^1_1(a), ..., \left(\vec{P}_2(a) - \overline{P}(a) \right), ..., P^1_{c_1}(a)) \right) + \overline{F}_1(a) \tag{4}$$

Now, we use the following key observation: $rank\left(\mathcal{P}_1\right) \geq r_{2.19}(\mathbb{F}, d, c_1)$, and as $\deg(\Gamma_1(P^1_1, ..., P^1_{c_1})) \leq d_F$ we can use Lemma 2.19 to achieve that Γ_1 is a polynomial of the form:

$$\Gamma_1(z_1, ..., z_{c_1}) = \sum_{\alpha \in [p-1]^{c_1}} C_\alpha \cdot \prod_{i=1}^{c_1} z_i^{\alpha_i} \tag{$\star$}$$

where $C_\alpha = 0$ whenever $\sum_{i=1}^{c_1}(\alpha_i \cdot \deg(P^1_i)) > d_F$.
Next, we substitute the polynomial structure of Γ_1 ($\star$) in the $\mathcal{P}_1$-decomposition of F (4), and observe what happens to each summand monomial with non-zero coefficients of Γ_1 in the expression after the substitution.
We will show that each such monomial is either $\mathcal{P}_2$-measurable, or a sum of a

$\mathcal{P}_2$-measurable function with a valid $\tilde{X}$-remainder, i.e. a polynomial of degree $\leq d_F$ that is $\equiv 0$ in $\tilde{X}$. Note that if this is true for each monomial, every linear combination of such monomials is also a sum of $\mathcal{P}_2$-measurable function with a valid $\tilde{X}$-remainder. Thus, this will also be true for the entire decomposition of F, as it is a linear combination of such monomials summed with a valid remainder $\overline{F}_1$. This will conclude the proof.
Let $\alpha = (\alpha_1, ..., \alpha_{c_1})$ be a vector of degrees that represents such a monomial. If $\alpha_{i^\star} = 0$, then the monomial is in the form:

$$\prod_{i\in[c_1]} {P_i}^{\alpha_i} = \prod_{i\in[c_1]\setminus\{i^\star\}} {P_i}^{\alpha_i}$$

and therefore it is clearly $\mathcal{P}_2$-measurable as all the polynomials in the expression above are in $\mathcal{P}_2$.
Next, if $\alpha_{i^\star} \neq 0$, then the monomial is in the form:

$$\prod_{i\in[c_1]} {P_i}^{\alpha_i} = (\vec{P_2} + \overline{P})^{\alpha_{i^\star}} \cdot \left(\prod_{i\in[c_1]\setminus\{i^\star\}} {P_i}^{\alpha_i} \right) \tag{5}$$

where $\sum_{i\in[c_1]}(\alpha_i \cdot \deg(P_i^1)) \leq d_F$. As $\deg(\vec{P_2} + \overline{P}) = \deg(P_{i^\star}) = d^\star$, we have:

$$\deg\left(\prod_{i\in[c_1]\setminus\{i^\star\}} {P_i}^{\alpha_i} \right) = \sum_{i\in[c_1]\setminus i^\star} (\alpha_i \cdot \deg(P_i^1)) \leq d_F - \alpha_{i^\star} \cdot d^\star$$

Now, we open the left brackets in (5), i.e $(\vec{P_2}+\overline{P})^{\alpha_{i^\star}}$. This enables us to separate the monomial to the part that only depend on $\vec{P_2}$ summed with a polynomial with bounded degree multiplied by $\overline{P}$ (and therefore a valid remainder). To be more specific, the monomial is in the form:

$$(\vec{P_2} + \overline{P})^{\alpha_{i^\star}} = {\vec{P_2}}^{\alpha_{i^\star}} + \overline{P_\alpha}$$

for some polynomial $\overline{P_\alpha}$ such that:

1. $\overline{P_\alpha}$ is of degree $\deg(\overline{P_\alpha}) \leq \max\left\{\deg(\vec{P_2}), \deg(\overline{P})\right\} \cdot \alpha_{i^\star} \leq \alpha_{i^\star} \cdot d^\star$
2. $\overline{P_\alpha}$ is a multiple of $\overline{P}$, and therefore $\overline{P_\alpha}|_{\tilde{X}} \equiv 0$

Therefore, by substituting the left brackets back to the equation (5) and as $\vec{P_2}$ and P_i for $i \neq i^\star$ are $\mathcal{P}_2$-measurable, one can see that the monomial is a sum of a $\mathcal{P}_2$-measurable polynomial with a valid remainder. Specifically, the remainder $\equiv 0$ in $\tilde{X}$, and its degree is $\leq \alpha_{i^\star} \cdot d^\star + d_F - \alpha_{i^\star} \cdot d^\star = d_F$. This concludes the proof of the claim. □

Now, it remains to prove the second part of the Theorem 5.6.

Claim 5.9. *If $\mathcal{B} \succeq_{syn} \bar{\mathcal{B}}$ for some polynomial factor $\bar{\mathcal{B}}$ with relative rank of at least $r(c') + c' + 1$ and rank of at least $r_{2.19}(\mathbb{F}, d, c') + c' + 1$, then we can require that $\mathcal{B}' \succeq_{syn} \bar{\mathcal{B}}$.*

Proof. We will show claim step-by-step. We denote by $\mathcal{P}, \bar{\mathcal{P}}, \mathcal{P}_1, \mathcal{P}_2$ the polynomial sets that generate the factors $\mathcal{B}, \bar{\mathcal{B}}, \mathcal{B}_1, \mathcal{B}_2$. Note that $\mathcal{B}_1, \mathcal{B}_2$ are the factors in the current step of the regularization process, and thus change in each step of the proof. We show that in each step, if $\mathcal{B} \succeq_{\text{syn}} \bar{\mathcal{B}}$ for some polynomial factor $\bar{\mathcal{B}}$ with relative rank of at least $r(c')+c'+1$ and rank of at least $r_{2.19}(\mathbb{F}, d, c')+c'+1$, then we can require that $\mathcal{B}_1 \succeq_{\text{syn}} \bar{\mathcal{B}}$, and also that $\mathcal{B}_2 \succeq_{\text{syn}} \bar{\mathcal{B}}$.
For the first part, we have $\mathcal{B}_1 \succeq_{\text{syn}} \bar{\mathcal{B}}$ by a simple usage of the second part of lemma 2.21, as:

$$rank\left(\bar{\mathcal{P}}\right) > r_{2.19}(\mathbb{F}, d, c') + c' + 1 \geq r_{2.19}(\mathbb{F}, d, c_1) + c_1 + 1$$

Now we prove the second part. We show that in the current regularization step, we could replace $P^1_{i^\star} \in \mathcal{P}_1$ such that $P^1_{i^\star} \notin \bar{\mathcal{P}}$. Note that this is possible whenever $\mathcal{P}^\star \cap \bar{\mathcal{P}} \neq \emptyset$ as the choice of $i^\star$ is arbitrary in polynomials which are in $\mathcal{P}^\star$.
Assume that is not possible and the factor $\mathcal{P}_1$ is still not r-$\tilde{X}$-regular. Then, we have a linear combination $\vec{P}(x) \coloneqq \sum_{i=0}^{c_1} \lambda_i P^1_i(x)$ with $rank_{d^\star,\tilde{X}}\left(\vec{P}\right) \leq r(c_1)$ where $d^\star = \max_{i\in[c_1]} \deg(\lambda_i P^1_i)$. We denote by $I^\star \subseteq [c_1]$ the set of indexes of such maximal-degree polynomials. By this notation, our assumption states that for all $i \in I^\star$ we have $P^1_i \in \bar{\mathcal{P}}$. Additionally, note that for all $i \notin I^\star$ we have $\deg(P^1_i) < d^\star$. Therefore, as the linear combination is of $d^\star$-relative rank $\leq r(c_1)$, there exists a polynomial $\overline{P}$ of degree $\leq \deg(\vec{P}) \leq d^\star$ with $\overline{P}|_{\tilde{X}} \equiv 0$ such that $rank_{d^\star}\left(\vec{P} - \overline{P}\right) \leq r(c_1)$. In other words, there exist a measurement function $\vec{\Gamma} : \mathbb{F}^{r(c_1)} \to \mathbb{F}$ and polynomials $Q_1, ..., Q_{r(c_1)}$ with $\deg(Q_i) \leq d^\star$ such that:

$$\forall a \in \mathbb{F}^n : \vec{P}(a) - \overline{P}(a) = \vec{\Gamma}\left(Q_1(a), ..., Q_{r(c_1)}(a)\right)$$

By a simple calculation we have:

$$\forall a \in \mathbb{F}^n : \sum_{i\in I^\star} P^1_i(a) - \overline{P}(a) = \vec{\Gamma}\left(Q_1(a), ..., Q_{r(c_1)}(a)\right) + \sum_{i\notin I^\star} P^1_i(a)$$

and by this we found a linear combination of polynomials in $\bar{\mathcal{P}}$ with maximal degree $d^\star$, that has $d^\star$-relative-rank $\leq r(c_1) + c_1 + 1$. This is a contradiction to our assumptions on $\bar{\mathcal{B}}$, which completes the proof of the claim. □

This completes the proof of the lemma. □

6 List Decoding Reed Muller Over $\tilde{X}$

In this section, we prove our main theorem: we prove the list decoding radius of Reed-Muller codes in is $\tilde{X}$ *at least* the list decoding radius of Reed-Muller codes in $\mathbb{F}^n$, assuming $\tilde{X}$ is lift-enabler and has the relative rank-bias property. We start by presenting formally the list decoding radius in $\tilde{X}$.

Definition 6.1 (List Decoding in $\tilde{X}$)**.** *Let $\mathbb{F}$ be a finite field. Let $d, n \in \mathbb{N}$, and let $\tilde{X} \subseteq \mathbb{F}^n$.*
We define the reed muller list-decoding count in $\tilde{X}$ at distance τ as follows:

$$\ell_{\mathbb{F},\tilde{X}}(d, \tau) \coloneqq \max_{F:\tilde{X}\to\mathbb{F}} \left|\left\{P \in Poly_{\leq d}(\tilde{X} \to \mathbb{F}) \middle| dist\left(P, F\right) \leq \tau\right\}\right|$$

Additionally, we define $LDR_{\mathbb{F},\tilde{X}}(d)$ *to be the* list decoding radius*, which is the maximum* τ *for which* $\ell_{\mathbb{F},\tilde{X}}(d, \tau - \epsilon)$ *is bounded by a* constant *depending only on* $\epsilon, |\mathbb{F}|, d$.

We recall that it was shown in [BL14, Theorem 1] that the list decoding radius of Reed Muller is $\delta_{\mathbb{F}}(d)$. To be more precise, it was shown that for every $\epsilon > 0$, the list-decoding count is constant (independent of n) in distance $\tau = \delta_{\mathbb{F}}(d) - \epsilon$. Formally, they have shown the following theorem:

Theorem 6.2 (List Decoding RM in $\mathbb{F}^n$)**.** *There exists a function* $c(\mathbb{F}, d, \epsilon)$ *such that the following holds: Let* $\mathbb{F}$ *be a finite field, let* $\epsilon > 0$*, and let* $d, n \in \mathbb{N}$*. Then, we have:*

$$\ell_{\mathbb{F},\mathbb{F}^n}(d, \delta_{\mathbb{F}}(d) - \epsilon) \leq c(\mathbb{F}, d, \epsilon)$$

Additionally, we recall a lemma that was presented in [BL14, Corollary 3.3], and was used in the analysis of the list decoding radius of Reed-Muller codes in $\mathbb{F}^n$:

Lemma 6.3 (Low Complexity Approximation)**.** *[BL14, Corollary 3.3] Let* $G : A \to B$*, and let* $\epsilon > 0$*. Let* $\mathfrak{F} \subseteq B^A$ *be a collection of functions from* A *to* B*. Then there exists* $c \leq 1/\epsilon^2$ *functions* $F_1, ..., F_c \in \mathfrak{F}$ *such that for every* $F \in \mathfrak{F}$*, there is a function* $\Gamma_F : B^c \to B$ *such that:*

$$\Pr_{x \in A}[\Gamma_F(F_1(x), ..., F_c(x)) = F(x)] \geq \Pr_{x \in A}[G(x) = F(x)] - \epsilon$$

The lemma shows that G can "estimated" by a only a few functions from $\mathfrak{F}$. Note that the estimation is close to G in compare to every $F \in \mathfrak{F}$ and not necessarily close to G itself.

Finally, we present our main theorem, which shows that under assumptions on the subset $\tilde{X} \subseteq \mathbb{F}^n$, the list decoding radius of polynomials in $\tilde{X}$ will be similar to the list decoding radius in $\mathbb{F}^n$.
In more details (and informally), we show that if $\tilde{X} \subseteq \mathbb{F}^n$ is lift-enabler, has the limited-relative-rank-bias-property, the list-decoding count is constant (independent of n) for every valid ϵ in distance $\tau = \delta_{\mathbb{F}}(d) - \epsilon$. Note that not every $\epsilon > 0$ will be valid: the valid values of ϵ will depend on the limitations of the rank-bias property. Formally, we show the following:

Theorem 6.4 (List Decoding RM in $\tilde{X}$)**.** *There exist functions* $c_1(\mathbb{F}, d, \tilde{r}, \tilde{\epsilon})$ *and* $c_2(\mathbb{F}, d, \tilde{r}, \epsilon)$ *such that the following holds: Let* $\mathbb{F}$ *be a finite field, and let* $d \in \mathbb{N}$ *be an integer that represents a degree. Let* $\tilde{\epsilon} > 0$*, and let* $\tilde{r} : [\tilde{\epsilon}, \infty] \to \mathbb{N}$ *be a limited-relative-rank-bias function.*
Let $\tilde{X} \subseteq \mathbb{F}^n$ *be a set with the following properties*

1. $\tilde{X}$ *is* d*-lift-enabler with a lift operator* $\widehat{\square}$*.*
2. $\tilde{X}$ *has the* $(\tilde{r}, \mathbb{F}, d, \tilde{\epsilon})$*-relative-rank-bias property.*

Then, for every $\epsilon \geq c_1(\mathbb{F}, d, \tilde{r}, \tilde{\epsilon})$ *it holds:*

$$\ell_{\mathbb{F},\tilde{X}}(d, \delta_{\mathbb{F},\mathbb{F}^n}(d) - \epsilon) \leq c_2(\mathbb{F}, d, \tilde{r}, \epsilon)$$

Proof. We follow the lines of the proof of [BL14, Theorem 1]. Let $\mathbb{F}$ be a finite field, and let $d \in \mathbb{N}$ be an integer that represents a degree. Let $\tilde{\epsilon} > 0$, and let

$\tilde{r} : [\tilde{\epsilon}, \infty] \to \mathbb{N}$ be a limited-rank-relative-bias function. Let $c_1(\mathbb{F}, d, \tilde{r}, \tilde{\epsilon})$ be a function we will specify later. Let $\tilde{X} \subseteq \mathbb{F}^n$ be a set with the properties defined above.
Finally, let $\epsilon \geq c_1(\mathbb{F}, d, \tilde{r}, \tilde{\epsilon})$ for c_1 that we will specify later, and let $f : \tilde{X} \to \mathbb{F}$ be a received word. We wish to bound the amount of polynomials in $Poly_{\leq d}(\tilde{X} \to \mathbb{F})$ that are $(\delta_{\mathbb{F}}(d) - \epsilon)$-close to f.
Apply Lemma 6.3 with $A = \tilde{X}$, $B = \mathbb{F}$, $G = f$, $\mathfrak{F} = Poly_{\leq d}(\tilde{X} \to \mathbb{F})$ and approximation parameter $\epsilon/2$ to obtain $\mathfrak{h} \subset Poly_{\leq d}(\tilde{X} \to \mathbb{F})$, defined by $\mathfrak{h} = (h_1, ..., h_c)$ where $c \leq 4/\epsilon^2$, such that for every $p \in Poly_{\leq d}(\tilde{X} \to \mathbb{F})$ there is a function $\Gamma_p : \mathbb{F}^c \to \mathbb{F}$ that approximates f in $\tilde{X}$ relative to $Poly_{\leq d}(\tilde{X} \to \mathbb{F})$ i.e.:

$$\forall p \in Poly_{\leq d}(\tilde{X} \to \mathbb{F}) : \Pr_{x \in \tilde{X}}[\Gamma_p(h_1(x), ..., h_c(x)) = p(x)] \geq \Pr_{x \in \tilde{X}}[f(x) = p(x)] - \epsilon/2$$

Let $r_1, r_2 : \mathbb{N} \to \mathbb{N}$ be two non-decreasing functions that represents rank that we will specify later. For r_1, we will require that for all $m \geq 1$:

$$r_1(m) \geq \max \left\{ \begin{matrix} r_2(C^{5.6}_{r_2,d}(m+1)) + C^{5.6}_{r_2,d}(m+1) + 1, \\ r_2(C^{5.6}_{r_{2.19},d}(m+1)) + C^{5.6}_{r_{2.19},d}(m+1) + 1 \end{matrix} \right\}$$

Note that in the expression above, we denote $r_{2.19} : \mathbb{N} \to \mathbb{N}$, as follows: $r_{2.19}(c) \coloneqq r_{2.19}(\mathbb{F}, d, c)$.
The reason we chose this r_1, is that by our choice of r_1 we can use the second part of Lemma 5.6. Specifically, if we start with r_1-$\tilde{X}$-regular factor and we r_2-$\tilde{X}$-regularize it, we get that the r_2-$\tilde{X}$-regular factor that we received is a syntactic refinement of the r_1-$\tilde{X}$-regular factor we started with.
As a first step, we lift the polynomial factor to get $\mathcal{H} \coloneqq \widehat{\mathfrak{h}}$. Note that because $\forall x \in \tilde{X} : \widehat{h_i}(x) = h_i(x)$, for all $p \in F$ we have:

$$\Pr_{x \in \tilde{X}}\left[\Gamma_p(\widehat{h_1}(x), ..., \widehat{h_c}(x)) = p(x)\right] \geq \Pr_{x \in \tilde{X}}[f(x) = p(x)] - \epsilon/2$$

Next, we r_1-$\tilde{X}$-regularize the factor $\mathcal{H}$ by Theorem 5.6. This gives us a r_1-$\tilde{X}$-regular factor $\mathcal{B}'$, which is defined by a set of polynomials $\mathcal{H}' \coloneqq (H'_1, ..., H'_{c'})$ of degree $\leq d$ such that $\mathcal{B}' \succeq^{\tilde{X}}_{\text{sem}} \mathcal{B}$, with $rank_{\tilde{X}}(\mathcal{H}') \geq r(c')$ and with bounded amount of polynomials defining it i.e. $c' \leq C^{5.6}_{r_1,d}(c)$. We apply Corollary 5.5 and get that $\mathcal{B}' \succeq_{\text{sem}|\tilde{X}} \mathcal{B}$. We then use the fact that $\Gamma_p(\widehat{h_1}(x), ..., \widehat{h_c}(x))$ is measurable in respect of $\mathcal{P}$ *in* $\tilde{X}$, and deduce we have a similar approximation of p using $\mathcal{P}'$ as the approximation of p using $\mathcal{P}$. Formally, there exists a function $\Gamma'_p : \mathbb{F}^{c'} \to \mathbb{F}$ such that:

$$\Pr_{x \in \tilde{X}}\left[\Gamma'_p(H'_1(x)), ..., H'_{c'}(x)) = p(x)\right] \geq \Pr_{x \in \tilde{X}}[f(x) = p(x)] - \epsilon/2$$

Now we recall that we wished to bound the amount of polynomials $p \in Poly_{\leq d}(\tilde{X} \to \mathbb{F})$ such that $\Pr_{x \in \tilde{X}}[f(x) \neq p(x)] < \delta_{\mathbb{F}}(d) - \epsilon$. Let $p \in Poly_{\leq d}(\tilde{X} \to \mathbb{F})$ be a polynomial as we just described. We will show that such p is measurable with respect to $\mathcal{H}'$ in $\tilde{X}$. This will upper bound the amount of possible polynomials p by the amount of possible different $\Gamma'_p : \mathbb{F}^{c'} \to \mathbb{F}$, which is $|\mathbb{F}|^{\|\mathcal{B}'\|} = p^{(p^{c'})}$,

and thus $c_2(\mathbb{F}, d, \tilde{r}, \epsilon) \leq p^{(p^{c'})}$.
By our choice of c' we have that $c' \leq C^{5.6}_{r_1,d}(4/\epsilon^2)$, and thus c_2 is bounded by a function of $(\mathbb{F}, d, r_1, \epsilon)$. Note that we have not yet specified the value of r_1, because it is determined by the choice of r_2 that we will later define its exact values. The important thing about our future choice of r_2 is that the value of r_2 must be independent of n, but can depend on $(\mathbb{F}, d, \tilde{r}, \epsilon)$. This will conclude the proof.
Now, consider a lift of p, i.e $P \coloneqq \widehat{p}$. Note that by the definition of lift $\forall x \in \tilde{X} : P(x) = p(x)$. We will show that P is measurable in respect of $\mathcal{H}'$ in $\tilde{X}$.
We consider the factor $\mathcal{B}_P$ that is generated by $\mathcal{H}_P \coloneqq \mathcal{H}' \cup \{P\}$. By using Theorem 5.6, we can r_2-$\tilde{X}$-regularize it and get the polynomial factor $\mathcal{B}''$ that relative-refines $\mathcal{B}_P$. We denote the set of polynomials in the factor as $\mathcal{H}''$.
Next, notice that the factor $\mathcal{B}''$ is a r_2-regular factor, therefore by our choice of r_1 and the second part of Theorem 5.6, we in fact have $\mathcal{B}'' \succeq_{\text{syn}} \mathcal{B}'$. This is true because by our choice of r_1:

$$rank_{\tilde{X}}(\mathcal{H}') \geq r_1(c') \geq r_2(C^{5.6}_{r_2,d}(c'+1)) + C^{5.6}_{r_2,d}(c'+1) + 1 \geq r_2(|\mathcal{B}''|) + |\mathcal{B}''| + 1$$

And as rank is always bigger than relative rank, we also have:

$$rank(\mathcal{H}') \geq r_1(c') \geq r_2(C^{5.6}_{r_{2.19},d}(c'+1)) + C^{5.6}_{r_{2.19},d}(c'+1) + 1$$

Thus, the polynomials defining $\mathcal{B}''$ are in the form $\mathcal{H}'' \coloneqq \mathcal{H}' \cup \{H''_1, ..., H''_{c''}\}$. Note that as promised in Theorem 5.6, we have $|\mathcal{H}''| = c' + c'' \leq C^{5.6_{r_2,d}}(c')$.
Additionally, by the way we built $\mathcal{H}_P$, the function P is measurable in respect of it. Therefore, as $\mathcal{B}'' \succeq^{\tilde{X}}_{\text{sem}} \mathcal{B}_P$, we have that P is $\mathcal{H}''$-measurable relative to $\tilde{X}$. In other words, there exists $\Phi : \mathbb{F}^{c'+c''} \to \mathbb{F}$ and $\overline{P} : \mathbb{F}^n \to \mathbb{F}$ with $\deg(\overline{P}), \deg(P - \overline{P}) \leq \deg(P) \leq d$ and $\overline{P}|_{\tilde{X}} \equiv 0$ such that:

$$\forall a \in \mathbb{F}^n : P(a) = \Phi(H'_1(a), ..., H'_{c'}(a), H''_1(a), ..., H''_{c''}(a))) + \overline{P}(a)$$

And specifically in $\tilde{X}$ we have:

$$\forall x \in \tilde{X} : P(x) = \Phi(H'_1(x), ..., H'_{c'}(x), H''_1(x), ..., H''_{c''}(x)))$$

Denote $P' \coloneqq P - \overline{P}$. We will show the polynomial P' does not depend on its last c'' variables, and thus Φ does not depend on its last c'' variables. This will imply that P is measurable in respect of $\mathcal{H}'$ in $\tilde{X}$, which will conclude the proof.
Now, we choose r_2 to be such that:

$$r_2(m) \geq \max\left\{\tilde{r}\left(\frac{\epsilon/4}{|\mathbb{F}|^m}\right), r_{2.12}\left(\frac{\epsilon/4}{|\mathbb{F}|^m}\right), r_{2.19}(m)\right\}$$

Note that in the expression above we are discussing fixed field and degree, i.e. $\mathbb{F}, d$. Therefore we denote $r_{2.19} : \mathbb{N} \to \mathbb{N}$ as $r_{2.19}(c) \coloneqq r_{2.19}(\mathbb{F}, d, c)$ and $r_{2.12} : \mathbb{N} \to \mathbb{N}$ as $r_{2.12}(\epsilon) \coloneqq r_{2.12}(\mathbb{F}, d, \epsilon)$.
Next, we show that even if we change the polynomials in the factor to have a disjoint set of inputs in $\mathbb{F}^n$, we still obtain a polynomial in the same degree, which have an approximation close to the approximation we had in $\tilde{X}$. Note that after this step, the proof becomes very similar to the proof of list decoding Reed Muller in $\mathbb{F}^n$ [BL14, Theorem 1]: we omit the dependence of $\tilde{X}$ and get the same approximation by functions of multiple variables, as we had in $\mathbb{F}^n$. This is done by the following lemma:

Lemma 6.5. *Let $\left\{a^i, b^j\right\}, i \in [c'], j \in [c'']$ be pairwise disjoint sets of n variables each. Let $n' \coloneqq n(c' + c'')$. Let $\vec{P} : \mathbb{F}^{n'} \to \mathbb{F}$ and $\vec{f} : \mathbb{F}^{n'} \to \mathbb{F}$ be functions of n' variables defined as follows:*

$$\vec{P'}(\vec{a}) \coloneqq \Phi\left(H'_1(a^1), ..., H'_{c'}(a^{c'}), H''_1(b^1), ..., H''_{c''}(b^{c''})\right)$$

and:

$$\vec{f}(\vec{a}) \coloneqq \Gamma'_p(H'_1(a^1)), ..., H'_{c'}(a^{c'}))$$

Note that $\vec{f}$ is a function that receives n' variables, and ignores its last c'' variables.
Then:

1. *The degree of $\vec{P'}$ remains bounded, i.e. $\deg(\vec{P'}) \leq d$.*
2. *The approximation of $\vec{f}$ to $\vec{P'}$ in $\mathbb{F}^{n'}$ is close to the approximation of Γ'_p to p in $\tilde{X}$. Specifically, we show:*

$$\left| \Pr_{\vec{a} \in \mathbb{F}^{n'}} \left[\vec{f}(\vec{a}) = \vec{P'}(\vec{a})\right] - \Pr_{x \in \tilde{X}} \left[\Gamma'_p(H'_1(x)), ..., H'_{c'}(x)) = p(x)\right] \right| \leq \epsilon/4$$

Proof. We start by proving the first part of the lemma: bounding the degree of $\vec{P'}$ by d. First, we recall that $P' = P - \overline{P}$ where $\overline{P}$ is a valid remainder. Specifically, we have $\deg(P') = \deg(P - \overline{P}) \leq \deg(P) \leq d$. In addition, by the way we built Φ we have:

$$\forall a \in \mathbb{F}^n : P'(a) = \Phi(H'_1(a), ..., H'_{c'}(a), H''_1(a), ..., H''_{c''}(a)))$$

Thus the function above is of degree $\leq d$. Moreover, we have:

$$rank\left(\mathcal{H}''\right) \geq rank_{\tilde{X}}\left(\mathcal{H}''\right) \geq r_2(|\mathcal{H}''|) \geq r_{2.19}(|\mathcal{H}''|)$$

Therefore we can use Lemma 2.19 to get that $\deg(\vec{P'}) \leq \deg(P') \leq d$. Note that in order to use the lemma formally, we had to extend the input space of P' to be of n' variables (and make it depend only on the first n variables as it used to). Because lemma 2.19 require bounds independent of n, this is done smoothly.
Now we move to the second part of the lemma: bounding the approximation of $\vec{f}$ to $\vec{P'}$. Denote $S \coloneqq \mathbb{F}^{c'+c''}$, and for each $s \in S$ denote:

$$p_1(s) \coloneqq \Pr_{x \in \tilde{X}} \left[(H'_1(x), ..., H'_{c'}(x), H''_1(x), ..., H''_{c''}(x)) = s\right]$$

and as of our choice of r_2, we have $rank\left(\mathcal{H}''\right) \geq \tilde{r}(\epsilon/8\,|S|)$. Therefore, if we require that the relative rank-bias relation holds for $\epsilon/8\,|S|$, we can use Lemma 7.2 with $A = \tilde{X}$ to get that p_1 is $(\epsilon/8\,|S|)$-almost uniform, i.e:

$$p_1(s) = \frac{1 \pm \epsilon/8}{|S|}$$

We show that this can be done in the following claim by choosing a proper c_1:

Claim 6.6. *One can choose $c_1 \coloneqq c_1(\mathbb{F}, d, \tilde{r}, \tilde{\epsilon})$ such that if $\epsilon \geq c_1$ we have that $\epsilon/8\,|S| \geq c_1$.*

Proof. This is done by using the bound we already know. We need that:

$$\tilde{\epsilon} \leq \frac{\epsilon}{8\left|\mathbb{F}\right|^{c'+c''}}$$

As $c'+c'' \leq C^{5.6}_{r_2,d}(c')$, for the term above to hold it is enough that the following will be true:

$$\epsilon \geq \tilde{\epsilon} \cdot 8\left|\mathbb{F}\right|^{C^{5.6}_{r_2,d}(c')}$$

and as r_2, c' and thus also $C^{5.6}_{r_2,d}(c')$ are independent of n, we can pick $c_1 = c_1(\mathbb{F}, d, \tilde{r}, \tilde{\epsilon})$ and get what we aimed for. □

Thus, we can assume that p_1 is $(\epsilon/8\left|S\right|)$-almost uniform. Now, let:

$$p_2(s) \coloneqq \Pr_{\vec{a}\in\mathbb{F}^{n'}}\left[\left(H'_1(a^1), ..., H'_{c'}(a^{c'}), H''_1(b^1), ..., H''_{c''}(b^{c''})\right) = s\right]$$

Note that the rank of $\vec{\mathcal{H}}'' = \left\{H'_1(a^1), ..., H'_{c'}(a^{c'}), H''_1(b^1), ..., H''_{c''}(b^{c''})\right\}$, as a factor defined over $\mathbb{F}^{n'}$, can not be lower than the rank of $\mathcal{H}''$ and thus we have $rank\left(\vec{\mathcal{H}}''\right) \geq r_{2.12}\left(\frac{\epsilon/8}{|\mathbb{F}|^m}\right)$. By using lemma 2.12, which shows the rank-bias relation for $\mathbb{F}^{n'}$, we can similarly use Lemma 7.2 with $A = \mathbb{F}^{n'}$ to get that p_2 is also $(\epsilon/8\left|S\right|)$-almost-uniform, i.e:

$$p_2(s) = \frac{1 \pm \epsilon/8}{|S|}$$

Now, we show the approximations are the same. Denote by s' the restriction of s to its first c' coordinates, and consider the approximation:

$$\begin{aligned}
\Pr_{\vec{a}\in\mathbb{F}^{n'}}\left[\vec{f}(\vec{a}) = \vec{P}'(\vec{a})\right] = & \\
&= \sum_{s\in S} p_2(s) \cdot 1_{\Phi(s)=\Gamma'_P(s')} \\
&= \sum_{s\in S} p_1(s) \cdot 1_{\Phi(s)=\Gamma'_P(s')} \pm \epsilon/4 \\
&= \Pr_{x\in\tilde{X}}\left[\Gamma'_p(H'_1(x)), ..., H'_{c'}(x)) = p(x)\right] \pm \epsilon/4
\end{aligned}$$

This completes the proof the lemma. □

The proof is followed by the same methods used in [BL14]. We repeat if for completeness. We next restate a lemma proved in [BL14, Claim 4.2], which is a varaiant of the Schwartz-Zippel lemma [Sch80, Zip79]:

Lemma 6.7. *Let d, n_1, $n_2 \in \mathbb{N}$ be integers. Let $P_1 \in Poly_{\leq d}(\mathbb{F}^{n_1+n_2} \to \mathbb{F})$, and let $F_1 : \mathbb{F}^{n_1} \to \mathbb{F}$ be a function. Assume the polynomial is $\delta_{\mathbb{F}}(d)$-close to the function, i.e:*

$$\Pr_{x_1,...,x_{n_1+n_2}\in\mathbb{F}}\left[P_1(x_1, ..., x_{n_1+n_2}) = F_1(x_1, ..., x_n)\right] > 1 - \delta_{\mathbb{F}}(d)$$

Then, P_1 does not depend on $x_{n_1+1}, ..., x_{n_1+n_2}$.

Now, apply Lemma 6.7 to $P_1 = \vec{P'}$, $F_1 = \vec{f}$, $n_1 = nc'$, $n_2 = nc''$. We obtain that $\vec{P'}$ does not depend on its last c'' variables, and thus by denoting $C_i \coloneqq H''_i(0)$ for $i \in [c'']$ we have:

$$\vec{P'}(\vec{a}) = \Phi\left(H'_1(a^1), ..., H'_{c'}(a^{c'}), C_1, ..., C_{c''}\right)$$

Now, for every $a \in \mathbb{F}^n$, if we substitute a in the i-th component of $\vec{a}$ for every $i \in [c']$ in the equation above, we get the following is true:

$$P'(a) = \Phi\left(H'_1(a), ..., H'_{c'}(a), C_1, ..., C_{c''}\right)$$

Hence P' does not depend on its last c'' variables. As explained earlier, this implies that P is measurable in respect of $\mathcal{H}'$ in $\tilde{X}$. This completes the proof of the theorem.

□

7 Equidistribution of Functions

Assume we have a collection of functions $(F_1, ..., F_c)$, where $F_i : A \to \mathbb{F}$ for some finite set A. We are interested in showing that the functions are equidistributed, which means that their values behave close to independent random variables. We begin by formulating this definition:

Definition 7.1 (Equidistribution of Functions)**.** *Given $\epsilon > 0$ and $A \subseteq \mathbb{F}^n$, we say a collection of functions $\mathfrak{F} = (F_1, ..., F_c)$ where $F_i : A \to \mathbb{F}$ is ϵ-equidistributed in A if for all $\vec{\alpha} = (\alpha_1, ..., \alpha_c) \in \mathbb{F}^c$ we have:*

$$\Pr_{x \in A}\left[(F_1(x), ..., F_c(x)) = \vec{\alpha}\right] = \frac{1}{|\mathbb{F}|^c} \pm \epsilon$$

The following is a standard lemma that shows that if every linear combination of a collection of functions has low bias, the collection is equidistributed. We repeat the steps of the proof of [HHL19, Lemma 7.24], but here, we think of A as any finite set (and not particularly $\mathbb{F}^n$):

Lemma 7.2. *Let $\epsilon > 0$, and let A be a finite set. Let $\mathfrak{F} = (F_1, ..., F_c)$ be a collection of functions defined over A, i.e. $F_i : A \to \mathbb{F}$. Assume each linear combination of the collection has low bias, i.e for each $\lambda = (\lambda_1, ..., \lambda_c) \in \mathbb{F}^c$ such that $\lambda \neq \vec{0}$ we have:*

$$bias_{x \in A}(\sum_{i=1}^{c} \lambda_i F_i) < \epsilon$$

Then, the collection $\mathfrak{F}$ is ϵ-equidistributed over A.
In particular, for $\epsilon < \frac{1}{|\mathbb{F}|^c}$, the lemma shows that each atom of $\mathfrak{F}$ is not empty i.e for all $\vec{\alpha}$ there is some $x \in A$ such that $(F_1(x), ..., F_c(x)) = \vec{\alpha}$.

Proof. We wish to show that for each $\vec{\alpha} \in \mathbb{F}^c$ we have:

$$\Pr_{x \in A}\left[(F_1(x), ..., F_c(x)) = \vec{\alpha}\right] = \frac{1}{|\mathbb{F}|} \pm \epsilon$$

We express the fraction of inputs that are in the atom $\vec{\alpha}$ the following way:

$$\Pr_{x\in A}\left[(F_1(x),...,F_c(x))\right] = \mathbb{E}_{x\in A}\left[\prod_{i=1}^{c} 1_{[F_i(x)=\alpha_i]}\right]$$

We use the fact that for every $0 \neq x \in \mathbb{F}$, we have $\sum_{\lambda=0}^{p-1} e\left[\lambda x\right] = 0$, and if $x = 0$ we have $\sum_{\lambda=0}^{p-1} e\left[\lambda x\right] = p$. Therefore, the expression above equals:

$$= \mathbb{E}_{x\in A}\left[\prod_{i=1}^{c}\left(\frac{1}{p}\cdot\sum_{\lambda_i=0}^{p-1} e\left[\lambda_i(F_i(x)-\alpha_i)\right]\right)\right] = \frac{1}{p^c}\cdot\mathbb{E}_{x\in A}\left[\prod_{i=1}^{c}\sum_{\lambda_i=0}^{p-1} e\left[\lambda_i(F_i(x)-\alpha_i)\right]\right]$$

By the definition of character functions, we have that $e\left[a+b\right] = e\left[a\right]\cdot e\left[b\right]$, and therefore the expression above equals:

$$\frac{1}{p^c}\cdot\sum_{(\lambda_1,...,\lambda_c)\in\prod_{i=1}^{c}[0,p-1]}\left(\mathbb{E}_{x\in A}\left[e\left[\sum_{i=0}^{c}\lambda_i(F_i(x)-\alpha_i)\right]\right]\right)$$

Now, we use the fact that:

$$bias_{x\in A}(\sum_{i=1}^{c}(\lambda_i(F_i(x)-\alpha_i)) = bias_{x\in A}(\sum_{i=1}^{c}(\lambda_i F_i(x))) < \epsilon$$

and get that:

$$\Pr_{x\in A}\left[(F_1(x),...,F_c(x)) = \vec{\alpha}\right] = \frac{1}{p^c}\cdot\left(1\pm\epsilon\prod_{i=1}^{c}p\right) = \frac{1}{|\mathbb{F}|^c}\pm\epsilon$$

□

8 Comparing Ranks

In this section, we compare the definition of rank we used in this paper to another definition of rank used implicitly throughout this paper. This comparison is crucial, as there is no universally accepted definition of rank; different theorems presented throughout this paper employ distinct definitions. We demonstrate that our definition is sufficiently comprehensive, in that a polynomial (or a factor) classified as having high rank according to our criteria also exhibits high rank according to the second implicitly-used definition. While in many cases the comparison may appear straightforward, we include it for the sake of completeness.

Specifically, we compare our definition of rank with the definition established in [LZ21]. The paper [LZ21] extended the original definition of rank that was presented in [Sch85], to include also the concept of relative rank. It is important to note that this definition is specifically defined to subsets $\tilde{X} \subseteq \mathbb{F}^n$ that can be expressed as sets in the form $\tilde{X} = Z\left(\tilde{\mathcal{L}}\right)$ for some set of polynomials $\tilde{\mathcal{L}}$, and not to a general set $\tilde{X} \subseteq \mathbb{F}^n$.

First, we present a useful notation that is used in the definition presented in [LZ21]:

Notation (Largest Degree Homogenous Part)**.** *For a polynomial P of degree d, we denote by* $\mathrm{h}(P)$ *its degree-d homogenous component. In other words,* $\mathrm{h}(P)$ *is the sum of all the monomials of P of degree exactly d. For a set of polynomials $\mathcal{P} = \{P_1, ..., P_c\}$, we define* $\mathrm{h}(\mathcal{P}) \coloneqq \{\mathrm{h}(P_i) | i = 1, ..., c\}$.

Next, we present the exact definition of rank for a polynomial:

Definition 8.1 (Schmidt Rank of a Polynomial)**.** *The schmidt rank of a homogenous polynomial $P : \mathbb{F}^n \to \mathbb{F}$, noted as schmrank (P), is the minimal r such that there exist $(Q_i, H_i)_{i\in[r]}$ with $\deg Q_i, \deg H_i < \deg P$ such that:*

$$P(x) = \sum_{i=1}^{r} (Q(x) \cdot H(x))$$

For a general polynomial P of degree d, we set its rank to be the rank of its degree-d homogenous component, i.e. schmrank $(P) \coloneqq$ schmrank $(\mathrm{h}(P))$.

Remark 8.2 (High rank implies high schmidt rank)**.** *If rank $(P) \geq 2 \cdot r + 1$ for some constant $r \in \mathbb{N}$, then schmrank $(P) \geq r$.*

Proof. For homogenous polynomial P, assume *schmrank* $(P) < r$. Then, there exist $r' < r$ such that there exist $(Q_i, H_i)_{i=1}^{r'}$ with $\deg Q_i, \deg H_i < \deg P$ such that:

$$P(x) = \sum_{i=1}^{r'} (Q(x) \cdot H(x))$$

Then we can choose $\Gamma : \mathbb{F}^{2r'} \to \mathbb{F}$ to be a sum of multiples of each two consecutive variables to get that $P(x) = \Gamma(Q_1(x), H_1(x), ..., Q_{r'}(x), H_{r'}(x))$, where the polynomials are from a degree $< \deg(P)$. This means that *rank* $(P) \leq 2r' < 2r$ as we requested.
If we do not assume P is homogenous, by adding $P - \mathrm{h}(P)$ as an input to Γ, one can create a $\Gamma' : \mathbb{F}^{2r'+1} \to \mathbb{F}$ which equals to P when substituting the inputs with some polynomials with degree $< \deg P$, which concludes the proof in a similar way. □

Next, we present the definition of Schmidt rank of a factor as defined in [LZ21].

Definition 8.3 (Schmidt Rank of a Factor)**.** *For a factor of homogenous polynomials $\mathcal{P} = (P_1, ..., P_c)$, the schmidt rank of the factor is defined as:*

$$\textit{schmrank}\,(\mathcal{P}) \coloneqq \min\left(\textit{schmrank}\left(\sum_{i=1}^{c} \lambda_i P_i \right) \middle| 0 \neq (\lambda_1, ..., \lambda_c) \in \mathbb{F}^c \right)$$

Similarly, for a factor of general polynomials $\mathcal{P}$, we set its rank to be the rank of its matching homogenous-factor, i.e. schmrank $(\mathcal{P}) \coloneqq$ schmrank $(\mathrm{h}(\mathcal{P}))$ For a factor $\mathcal{B}$ generated by $\mathcal{P}$, we define schmrank $(\mathcal{B}) \coloneqq$ schmrank $(\mathcal{P})$.

To establish the equivalence of this definition with the one employed throughout the paper, we must first acknowledge two key distinctions between the definitions. The first distinction is that this definition focuses on the largest-degree homogeneous components of the polynomials involved in the factor, rather than

considering linear combinations of polynomials from the factor. The second distinction pertains to the treatment of d in the computation of d-rank of each linear combination. This definition uses the degree of the linear combination directly to calculate the rank that participates in the minimum, in contrast to our definition which uses $\max_i \deg(\lambda_i P_i)$. Despite these differences, we will demonstrate that both definitions ultimately yield a similar rank assessment, thereby affirming their equivalence.

Remark 8.4 (High Rank Implies High Schmidt Rank for Factors)**.** *Let $\mathcal{P} = (P_1, ..., P_c)$ be a set of polynomials and let $r \in \mathbb{N}$ be a positive integer, i.e. $r > 0$. If $rank(\mathcal{P}) \geq 2 \cdot r + 1$, then $schmrank(\mathcal{P}) \geq r$.*

Proof. Assume that $schmrank(\mathcal{P}) \leq r$ for $r > 0$. We will show that $rank(\mathcal{P}) \leq 2r + 1$. By definition, there exists a linear combination of polynomials in $\mathrm{h}(\mathcal{P})$ with rank $\leq r$. In other words, there exists $\vec{0} \neq \lambda \in \mathbb{F}^c$ such that $schmrank(\sum_{i=1}^{c} \lambda_i \mathrm{h}(P_i)) \leq r$. Denote $\vec{P_h} \coloneqq \sum_{i=1}^{c} \lambda_i \mathrm{h}(P_i)$. As was shown in a previous remark, a rank of a polynomial is smaller than its schmidt rank up to a constant factor, thus $rank\left(\vec{P_h}\right) \leq 2r + 1$ (see Remark 8.2).

Next, we denote $\vec{P} \coloneqq \sum_{i=1}^{c} \lambda_i P_i$, and $d_M \coloneqq \max_{i \in [c]} \lambda_i P_i$. Note that $\deg(\vec{P}) \leq d_M$. We wish to show that $rank_{d_M}\left(\vec{P}\right) \leq 2r + 1$. First, we observe that the d_M-degree homogenous component of $\vec{P_h}$ equals the d_M-degree homogenous component of $\vec{P}$. This is true because every highest-degree component of polynomials in the linear combination that generated $\vec{P}$, also exists in the linear combination that generates $\vec{P_h}$. In particular, all homogenous components of degree d_M exists in both linear combinations $\vec{P_h}$ and $\vec{P}$. Therefore, if the degree of $\vec{P}$ equals d_M, we have that $rank_{d_M}\left(\vec{P}\right) = rank\left(\vec{P}\right) \geq 2r + 1$. Otherwise, if $\deg(\vec{P}) < d_M$, then $rank_{d_M}\left(\vec{P}\right) = 1 \leq 2r + 1$. This completes the proof. □

Note. *In the case discussed above, if $\deg(\vec{P}) < d_M$, then $schmrank(\mathcal{P}) = 0$.*

Proof. Assume that $\deg(\vec{P}) < d_M$. Therefore, the degree of the linear combination $\vec{P} = \sum_{i=1}^{c} \lambda_i P_i$ is strictly smaller than the degree of at least one of the polynomials participating in it. Denote by $\vec{\lambda}^\star$ the sub-combination of $\vec{\lambda}$ that consists only the polynomials that participated in $\vec{P}$ that are of degree $= d_M$. Trivially, $\vec{\lambda}^\star \neq \vec{0}$. Additionally, we have $\deg(\sum_{i=1}^{c} \lambda_i^\star P_i) < d_M$. Now, we use the following observation: the linear combination above, when summing only the homogenous components of each polynomial, equals 0, i.e. $\sum_{i=1}^{c} \lambda_i^\star \mathrm{h}(P_i) \equiv 0$. By this, we found a linear combination of $\mathrm{h}(\mathcal{P})$ that is $\equiv 0$. Thus by definition, we have $schmrank(\mathcal{P}) = 0$. □

Note. *This shows that if we compare only the differences in the definition of rank of a factor, i.e. the focus on linear combinations of the largest-degree homogenous components in contrast to the use of the maximal degree d-rank, the two definitions for a rank of a factor are equal up to ± 1 (in case we use the same definition of rank for a single polynomial). To avoid confusion, we omit the exact definitions and respective proof.*

We now present the definition of relative rank as stated in [LZ21, Definition 1.6]: We remind the reader that this definition is specifically defined to subsets $\tilde{X} \subseteq \mathbb{F}^n$ that can be expressed by $\tilde{X} = Z\left(\tilde{\mathcal{L}}\right)$ for some set of polynomials $\tilde{\mathcal{L}}$, and not to a general set $\tilde{X} \subseteq \mathbb{F}^n$.

Definition 8.5 (Relative Schmidt Rank of a Polynomial)**.** *The relative schmidt rank of a homogeneous polynomial P relative to a collection of homogeneous polynomials $\tilde{\mathcal{L}} = (L_1, \ldots, L_{\tilde{c}})$ is*

$$shcmrank_{\tilde{\mathcal{L}}}(P) \coloneqq \min\left\{ schmrank\left(P + \sum_{i=1}^{\tilde{c}} R_i L_i\right) \middle| \deg(L_i) + \deg(R_i) \leq \deg(P), \forall i \in [\tilde{c}] \right\}$$

Note that whenever $\deg L_i > \deg P$, this implies $R_i = 0$.
For general polynomial P and general collection of polynomials $\tilde{\mathcal{L}}$, we define the schmidt rank of the former in respect to the latter by the relative rank of their largest-degree homogenous component, i.e. $shcmrank_{\tilde{\mathcal{L}}}(P) \coloneqq shcmrank_{\mathrm{h}(\tilde{\mathcal{L}})}(\mathrm{h}(P))$.

Remark 8.6 (High Relative Rank $\Rightarrow$ High Relative Schmidt Rank)**.** *Let P and $\tilde{\mathcal{L}} = \{L_1, ..., L_{\tilde{c}}\}$ be polynomials, and let $\tilde{X} \subseteq \mathbb{F}^n$ be defined as $\tilde{X} = Z\left(\tilde{\mathcal{L}}\right)$.*

If $rank_{\tilde{X}}(P) \geq 2 \cdot r + 2$ for some constant $r \in \mathbb{N}$, then $shcmrank_{\tilde{\mathcal{L}}}(P) \geq r$.

Proof. Let P and $L_1, ..., L_{\tilde{c}}$ be polynomials. Assume that $shcmrank_{\tilde{\mathcal{L}}}(P) \leq r$. Then, there exists $R_1, ..., R_{\tilde{c}}$ with $\deg(L_i) + \deg(R_i) \leq \deg(P)$ for all $i \in [\tilde{c}]$ such that:

$$schmrank\left(\mathrm{h}(P) + \sum_{i=1}^{\tilde{c}} R_i \mathrm{h}(L_i)\right) \leq r$$

Denote $\overline{P_h} \coloneqq \sum_{i=1}^{\tilde{c}} R_i \mathrm{h}(L_i)$. As we have shown earlier, a rank of a polynomial is smaller than its schmidt rank up to a constant factor (See Remark 8.2). Thus:

$$rank\left(\mathrm{h}(P) + \overline{P_h}\right) \leq 2 \cdot schmrank\left(\mathrm{h}(P) + \overline{P_h}\right) + 1 \leq 2 \cdot shcmrank_{\tilde{X}}(P) + 1 = 2r + 1$$

Next, we denote the respective remainder polynomial for the non-homogenous analogue, i.e $\overline{P} \coloneqq \sum_{i=1}^{\tilde{c}} R_i L_i$. By observing the highest degree homogenous component of each summand, one can see that $\mathrm{h}(P + \overline{P}) = \mathrm{h}(\mathrm{h}(P) + \overline{P_h})$. Therefore, by adding to the decomposition the non higest-degree-homogenous-component, one can see that:

$$rank\left(P + \overline{P}\right) \leq rank\left(\mathrm{h}(P) + \overline{P_h}\right) + 1 \leq 2r + 2$$

This completes the proof as $rank_{\tilde{X}}(P) \leq rank\left(P + \overline{P}\right) \leq 2r + 2$. □

Remark 8.7 (Relative Schmidt Rank over Varieties of High Degree)**.** *If the polynomials defining the variety $\tilde{\mathcal{L}} = (L_1, ..., L_{\tilde{c}})$ are of degree $> \deg(P)$, then, $shcmrank_{\tilde{\mathcal{L}}}(P) = schmrank(P)$. This is true because in this case, in the calculation of the minimum in the definition of relative schmidt rank, we must have $R_i = 1$ for all $i \in [\tilde{c}]$ and therefore the minimum above is simply $rank(P)$. Note that a similar statement holds for factors aswell. If $\mathcal{P} = (P_1, ..., P_c)$ is a factor of degree d, then if all the polynomials in $\tilde{\mathcal{L}}$ have degree $> d$, then the statement above is also true i.e $shcmrank_{\tilde{\mathcal{L}}}(\mathcal{P}) = schmrank(\mathcal{P})$. This is true because for every linear combination of $\mathcal{P}$ has degree $\leq d$ and therefore its relative schimdt rank equals its rank.*

Finally, we present the extension of the definition of relative rank for polynomials factors:

Definition 8.8 (Relative Schmidt Rank of a Factor)**.** *The relative rank of a set of homogenous polynomials $\mathcal{P} = \{P_1, ..., P_c\}$ relative to another collection of polynomials $\tilde{\mathcal{L}} = \{L_1, ..., L_{\tilde{c}}\}$ is defined as:*

$$shcmrank_{\tilde{\mathcal{L}}}(\mathcal{P}) \coloneqq \min\left\{ shcmrank_{\tilde{\mathcal{L}}}\left(\sum_{i=1}^{c} \lambda_i P_i\right) \middle| \vec{0} \neq (\lambda_1, ..., \lambda_c) \in \mathbb{F}^c \right\}$$

If $\mathcal{P}$ is a general collection of polynomials, then $shcmrank_{\tilde{\mathcal{L}}}(\mathcal{P}) \coloneqq shcmrank_{\tilde{\mathcal{L}}}(\mathrm{h}(\mathcal{P}))$. For a factor $\mathcal{B}$ generated by a set of polynomials $\mathcal{P}$, we define its schmidt rank relative to $\tilde{X} = Z\left(\tilde{\mathcal{L}}\right)$ to be $shcmrank_{\tilde{X}}(\mathcal{B}) \coloneqq shcmrank_{\tilde{\mathcal{L}}}(\mathcal{P})$.

Remark 8.9. *Let $\mathcal{P} = \{P_1, ..., P_c\}$ and $\tilde{\mathcal{L}} = \{L_1, ..., L_{\tilde{c}}\}$ be sets of polynomials, and let $\tilde{X} \subseteq \mathbb{F}^n$ be defined as $\tilde{X} = Z\left(\tilde{\mathcal{L}}\right)$. Additionally, let $r \in \mathbb{N}$ such that $r > 0$. If $rank_{\tilde{X}}(\mathcal{P}) \geq 2 \cdot r + 2$ for some constant $r \in \mathbb{N}$, then $shcmrank_{\tilde{\mathcal{L}}}(P) \geq r$.*

Proof. Assume that $shcmrank_{\tilde{\mathcal{L}}}(\mathcal{P}) \leq r$. We will show that $rank_{\tilde{X}}(\mathcal{P}) \leq 2r + 2$. Let $\vec{0} \neq \vec{\lambda} \in \mathbb{F}^c$ be some vector of coefficients. Let $\vec{P} \coloneqq \sum_{i=1}^{c} \lambda_i P_i$ and $\vec{P_h} \coloneqq \sum_{i=1}^{c} \lambda_i \mathrm{h}(P_i)$ be the linear combinations of polynomials in $\mathcal{P}$ and $\mathrm{h}(\mathcal{P})$ with coefficients $\vec{\lambda}$ respectively, and let $d_M \coloneqq \max_{i \in [c]} \deg(\lambda_i P_i)$. Additionally, denote $\hat{r} \coloneqq shcmrank_{\tilde{\mathcal{L}}}\left(\vec{P_h}\right) \leq r$. It is enough to show that $rank_{d_M, \tilde{X}}\left(\vec{P}\right) \leq 2\hat{r} + 2$, If $\deg(\vec{P}) < d_M$, then $rank_{d_M, \tilde{X}}\left(\vec{P}\right) = 1 \leq 2r + 2$. Otherwise, if $\deg(\vec{P}) = d_M$, then the remark follows from Remark 8.6 as:

$$rank_{d_M, \tilde{X}}\left(\vec{P}\right) = rank_{\tilde{X}}\left(\vec{P}\right) \leq 2 \cdot shcmrank_{\tilde{\mathcal{L}}}\left(\vec{P}\right) + 2$$

Where:

$$shcmrank_{\tilde{\mathcal{L}}}\left(\vec{P}\right) shcmrank_{\tilde{\mathcal{L}}}\left(\mathrm{h}(\vec{P})\right) = shcmrank_{\tilde{\mathcal{L}}}\left(\mathrm{h}(\vec{P_h})\right) = shcmrank_{\tilde{\mathcal{L}}}\left(\vec{P_h}\right) = \hat{r}$$

□

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